

Mathematics – I (M-I)
GTU # 3110014



Dedicated Faculty, Committed Education

Darshan
Institute of Engineering & Technology

Unit – 5

Multiple integral



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Sr. No.	Content	Total Hrs	% Weightage
05	Multiple integral, Double integral over Rectangles and general regions, double integrals as volumes, Change of order of integration, double integration in polar coordinates, Area by double integration, Triple integrals in rectangular, cylindrical and spherical coordinates, Jacobian, multiple integral by substitution.	08	20 %

Unit 5 Multiple Integral

- ✓ **Double integral over Rectangles and general regions**
- ✓ Double integrals as volumes
- ✓ Change of order of integration
- ✓ Double integration in polar coordinates
- ✓ Area by double integration
- ✓ **Triple integrals in rectangular**
- ✓ Cylindrical and spherical coordinates
- ✓ Jacobian
- ✓ Multiple integral by substitution

Outline

- ➡ Introduction
- ➡ Double Integrals by Direct Integration
- ➡ Triple Integrals by Direct Integration
- ➡ Double Integrals over General Region - CC

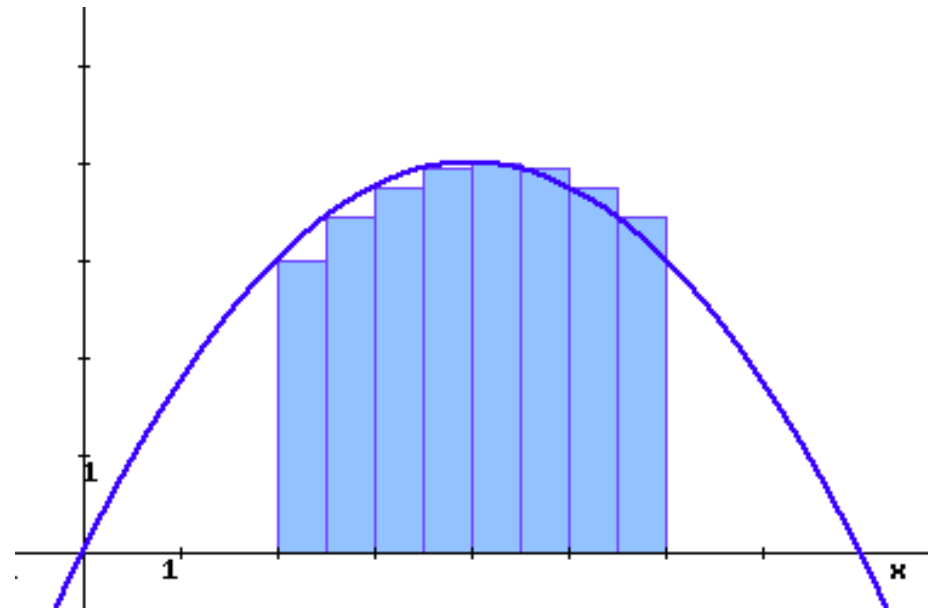


Introduction



What is Integral?

- ▶ The integral is usually called the **anti-derivative**, because **integrating** is the reverse process of **differentiating**.
- ▶ Integration can be **used to find** areas, volumes, central points and many useful things.
- ▶ Integration is a way of **adding slices** to find the **whole**.



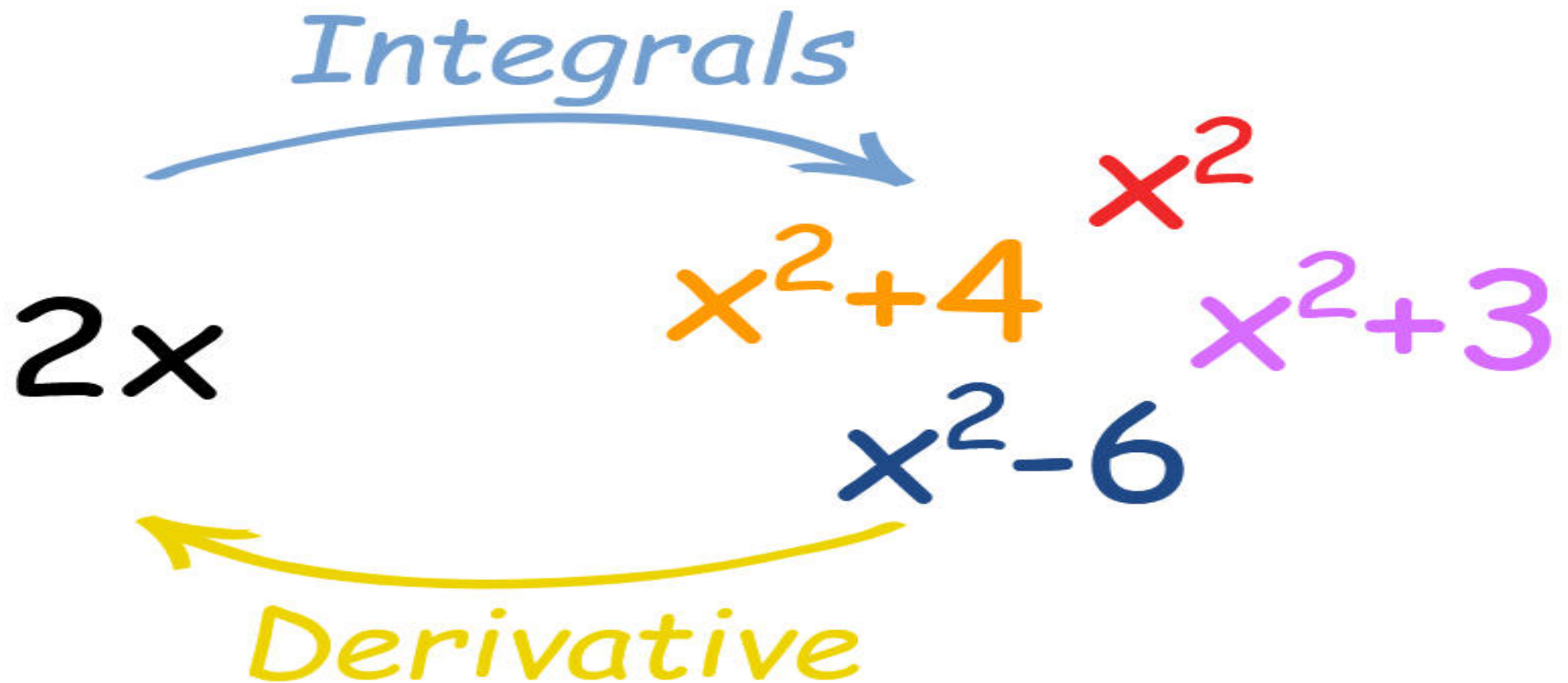
Slices along x

$$\int 2x \, dx$$

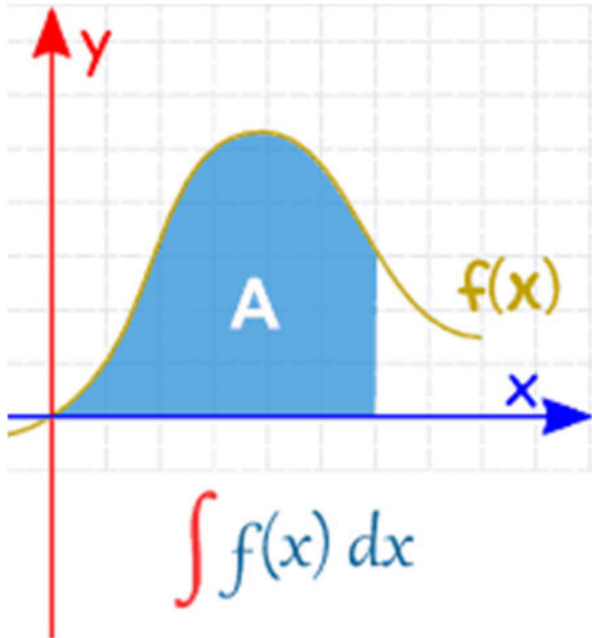
Integral Symbol

Function we want to integrate

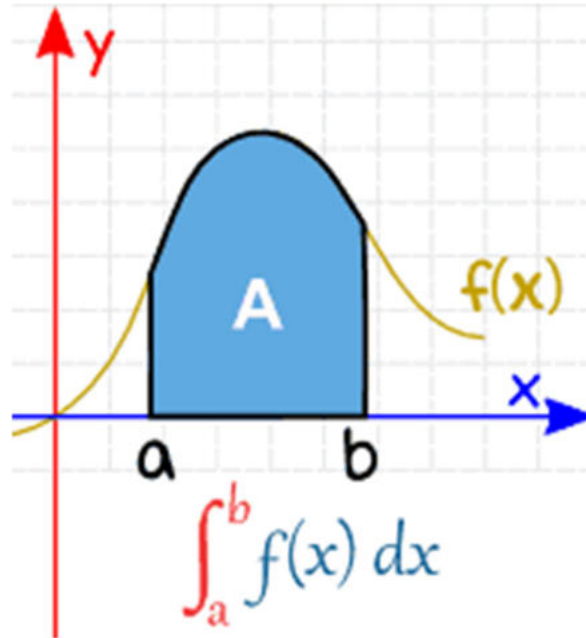
What is +c in Integral?



Definite vs Indefinite Integrals



Indefinite Integral



Definite Integral



Double Integrals By Direct Integration

What is Multiple Integral?

- ▶ The multiple integral is a **definite** integral of a **function of more than one** real variable.

For example: $f(x, y)$ or $f(x, y, z)$

- ▶ We will **study** Double integrals & Triple integrals.

What is Double Integral?

► Integrals of a function of **two variables** over a region in $\mathbb{R}^2$ are called **double integrals**.

► For example:

Double Integral of a function of two variables **$f(x, y)$** over region R in $\mathbb{R}^2$ is

$$\iint_R f(x, y) \, dA = \iint_R f(x, y) \, dx \, dy = \iint_R f(x, y) \, dy \, dx$$

Double Integral with Constant Limit

$$\int_{y=a}^{y=b} \left(\int_c^d f(x, y) \, dx \right) dy$$

=

$$\int_{x=c}^{x=d} f(x, y) \, dx = g(y)$$

Double Integral with Constant Limit

$$\int_a^b \int_c^d f(x, y) \, dx \, dy = \int_{y=a}^{y=b} \left(\int_{x=c}^{x=d} f(x, y) \, dx \right) dy$$

Double Integral with Constant Limit

$$\int_{x=a}^{x=b} \int_c^d f(x, y) \, dy \, dx$$

=

$$\int_{y=c}^{y=d} f(x, y) \, dy = g(x)$$

Double Integral with Constant Limit

$$\int_a^b \int_c^d f(x, y) \, dy \, dx = \int_{x=a}^{x=b} \left(\int_{y=c}^{y=d} f(x, y) \, dy \right) dx$$

Method 1 $\Rightarrow$ Example – 3

Evaluate $\int_0^1 \int_1^2 xy \, dy \, dx$.

Solution: W-19 (3 marks)

$$\begin{aligned} I &= \int_0^1 \int_1^2 xy \, dy \, dx \\ &= \int_0^1 \left(\int_{y=1}^{y=2} xy \, dy \right) dx \end{aligned}$$

$$\begin{aligned} &= \int_0^1 \left[x \cdot \frac{y^2}{2} \right]_{y=1}^{y=2} dx \\ &= \int_0^1 \left[\frac{x}{2} \cdot (4 - 1) \right] dx \\ &= \frac{3}{2} \int_0^1 [x] \, dx \\ I &= \frac{3}{4} \end{aligned}$$

.....

Method 1 $\Rightarrow$ Example – 5

Evaluate $\int_1^2 \int_0^1 \frac{1}{(2x + 5y)^2} dx dy$.

Solution:

$$I = \int_1^2 \int_0^1 \frac{1}{(2x + 5y)^2} dx dy$$

$$= \int_1^2 \left[\int_{x=0}^{x=1} (2x + 5y)^{-2} dx \right] dy$$

$$= \int_1^2 \left[\frac{(2x + 5y)^{-1}}{-1 \cdot 2} \right]_{x=0}^{x=1} dy$$

$$= \frac{-1}{2} \int_1^2 (2 + 5y)^{-1} - (0 + 5y)^{-1} dy$$

$$= \frac{-1}{2} \int_1^2 \frac{1}{2 + 5y} - \frac{1}{5y} dy$$

$$\int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)}$$

Method 1 $\Rightarrow$ Example – 5(continue)

$$I = \frac{-1}{2} \int_1^2 \frac{1}{2+5y} - \frac{1}{5y} dy$$

$$= \frac{-1}{2} \left[\left(\frac{1}{5} \log(2+5y) \right) - \left(\frac{1}{5} \log(y) \right) \right]_{y=1}^{y=2}$$

$$= \frac{-1}{10} [(\log 12 - \log 2) - (\log 7 - \log 1)]$$

$$I = 0.1 [-\log 12 + \log 2 + \log 7]$$

$$\int \frac{dx}{(ax+b)} \quad \int \frac{1}{x} dx = \log |x|$$

Method 1 $\Rightarrow$ Example – 6

Evaluate $\int_0^1 \int_0^1 \frac{1}{\sqrt{(1-x^2)(1-y^2)}} dx dy$.

Solution:

$$I = \int_0^1 \int_0^1 \frac{1}{\sqrt{(1-x^2)(1-y^2)}} dx dy = \int_0^1 \int_0^1 \frac{1}{\sqrt{(1-x^2)} \cdot \sqrt{(1-y^2)}} dx dy$$

$$= \int_0^1 \left(\int_{x=0}^{x=1} \frac{1}{\sqrt{(1-y^2)}} \cdot \frac{1}{\sqrt{(1-x^2)}} dx \right) dy$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a}$$

Method 1 $\Rightarrow$ Example – 6(continue)

$$I = \int_0^1 \frac{1}{\sqrt{(1-y^2)}} \left[\sin^{-1}(x) \right]_{x=0}^{x=1} dy$$

$$= \int_0^1 \frac{1}{\sqrt{(1-y^2)}} \left[\frac{\pi}{2} - 0 \right] dy$$

$$= \frac{\pi}{2} \int_0^1 \frac{1}{\sqrt{(1-y^2)}} dy$$

.....

$$I = \frac{\pi^2}{4}$$

Method 1 $\Rightarrow$ Example – 8

Evaluate $\int_1^4 \int_{2x^2}^{3x^2} (x e^{x^2+y}) dy dx$.

Solution:

$$\begin{aligned} I &= \int_1^4 \int_{2x^2}^{3x^2} (x e^{x^2+y}) dy dx \\ &= \int_1^4 \left(\int_{y=2x^2}^{y=3x^2} (x \cdot e^{x^2} \cdot e^y) dy \right) dx \end{aligned}$$

$$I = \int_1^4 x \cdot e^{x^2} \left[e^y \right]_{y=2x^2}^{y=3x^2} dx$$

$$I = \int_1^4 x \cdot e^{x^2} (e^{3x^2} - e^{2x^2}) dx$$

$$I = \int_1^4 (x \cdot e^{4x^2} - x \cdot e^{3x^2}) dx$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)}$$

Method 1 $\Rightarrow$ Example – 8(continue)

$$I = \int_1^4 \left(\frac{8x}{8} \cdot e^{4x^2} - \frac{6x}{6} \cdot e^{3x^2} \right) dx$$

$$= \left[\frac{e^{4x^2}}{8} - \frac{e^{3x^2}}{6} \right]_{x=1}^{x=4}$$

.....

$$I = \frac{e^{64} - e^4}{8} - \frac{e^{48} - e^3}{6}$$

Method 1 $\Rightarrow$ Example - 13

Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{1-\sin \theta} r^2 \cos \theta \, dr \, d\theta$.

Solution:

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \int_0^{1-\sin \theta} r^2 \cos \theta \, dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \left(\int_{r=0}^{r=1-\sin \theta} r^2 \cos \theta \, dr \right) d\theta \end{aligned}$$

$$= \int_0^{\frac{\pi}{2}} \cos \theta \left[\frac{r^3}{3} \right]_{r=0}^{r=1-\sin \theta} d\theta$$

$$= \frac{1}{3} \int_0^{\frac{\pi}{2}} \cos \theta [(1 - \sin \theta)^3] d\theta$$

$$= \frac{-1}{3} \int_0^{\frac{\pi}{2}} (-\cos \theta) (1 - \sin \theta)^3 d\theta$$

$$\int [f(x)]^n f'(x) \, dx = \frac{[f(x)]^{n+1}}{n+1}, n \neq -1$$

Method 1 $\Rightarrow$ Example – 13(continue)

$$I = \frac{-1}{3} \int_0^{\frac{\pi}{2}} (-\cos \theta) (1 - \sin \theta)^3 d\theta$$

$$I = \frac{-1}{3} \left[\frac{(1 - \sin \theta)^4}{4} \right]_{\theta=0}^{\theta=\frac{\pi}{2}}$$

$$I = \frac{1}{12}$$

.....



Triple Integrals By Direct Integration

What is Triple Integral?

► Integrals of a function of **three variables** over a region in $\mathbb{R}^3$ are called **triple integrals**.

► For example:

Double Integral of a function of three variables **$f(x, y, z)$** over region R in $\mathbb{R}^3$ is

$$\iiint_R f(x, y, z) \, dV = \iiint_R f(x, y, z) \, dx \, dy \, dz = \iiint_R f(x, y, z) \, dy \, dx \, dz = \dots$$

Triple Integral with Constant Limit

$$\int_m^n f(x, y, z) \, dz$$

$$\int_{z=m}^{z=n} f(x, y, z) \, dz = g(x, y)$$

=

Triple Integral with Constant Limit

$$\int_m^n f(x, y, z) \, dx$$

$$\int_{x=m}^{x=n} f(x, y, z) \, dx = g(y, z)$$

=

Triple Integral with Constant Limit

$$\int_a^b \int_c^d \int_m^n f(x, y, z) \, dx \, dy \, dz = \int_{z=a}^{z=b} \int_{y=c}^{y=d} \left(\int_{x=m}^{x=n} f(x, y, z) \, dx \right) dy \, dz$$

$$= \int_{z=a}^{z=b} \left(\int_{y=c}^{y=d} g(y, z) \, dy \right) dz = \int_{z=a}^{z=b} h(z) \, dz$$

Triple Integral with Constant Limit

$$\int_a^b \int_c^d \int_m^n f(x, y, z) \, dz \, dx \, dy = \int_{y=a}^{y=b} \int_{x=c}^{x=d} \left(\int_{z=m}^{z=n} f(x, y, z) \, dz \right) dx \, dy$$

$$= \int_{y=a}^{y=b} \left(\int_{x=c}^{x=d} g(x, y) \, dx \right) dy = \int_{y=a}^{y=b} h(y) \, dy$$

Method 2 $\Rightarrow$ Example – 2

Evaluate $\int_{-1}^1 \int_0^2 \int_0^1 xz - y^3 \, dz \, dy \, dx$.

Solution:

$$\begin{aligned} I &= \int_{-1}^1 \int_0^2 \int_0^1 xz - y^3 \, dz \, dy \, dx \\ &= \int_{-1}^1 \int_0^2 \left[\int_0^1 xz - y^3 \, dz \right] dy \, dx \end{aligned}$$

$$= \int_{-1}^1 \int_0^2 \left[x \frac{z^2}{2} - y^3 z \right]_{z=0}^{z=1} dy \, dx$$

$$= \int_{-1}^1 \int_0^2 \left[\left(\frac{x}{2} - y^3 \right) - (0) \right] dy \, dx$$

$$= \int_{-1}^1 \int_0^2 \frac{x}{2} - y^3 \, dy \, dx$$

Method 2 $\Rightarrow$ Example – 2(continue)

$$I = \int_{-1}^1 \left[\int_0^2 \frac{x}{2} - y^3 \, dy \right] dx$$

$$= \int_{-1}^1 \left[\frac{x}{2} y - \frac{y^4}{4} \right]_{y=0}^{y=2} dx$$

$$= \int_{-1}^1 [(x - 4) - (0)] dx$$

$$= \int_{-1}^1 [x - 4] dx$$

.....

$$I = -8$$

Method 2 $\Rightarrow$ Example – 3

Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx$.

Solution:

$$\begin{aligned} I &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx \\ &= \int_0^1 \int_0^{1-x} \left[\frac{z^2}{2} \right]_{z=0}^{z=1-x-y} dz \, dy \, dx \end{aligned}$$

$$\begin{aligned} I &= \frac{1}{2} \int_0^1 \int_0^{1-x} (1-x-y)^2 \, dy \, dx \\ &= \frac{1}{2} \int_0^1 \left[\frac{(1-x-y)^3}{3 \cdot (-1)} \right]_{y=0}^{y=1-x} dx \\ &= \frac{-1}{6} \int_0^1 [(0) - (1-x)^3] dx \end{aligned}$$

$$\int (ax + b)^n \, dx = \frac{(ax + b)^{n+1}}{a(n+1)}$$

Method 2 $\Rightarrow$ Example – 3(continue)

$$I = \frac{-1}{6} \int_0^1 [-(1-x)^3] dx$$

$$= \frac{+1}{6} \int_0^1 (1-x)^3 dx$$

.....

$$I = \frac{1}{24}$$

Integration by Parts

$$\int (\mathbf{u} * \mathbf{v}) \, dx = \mathbf{u} * \int \mathbf{v} \, dx$$

$$- \int \left\{ \frac{du}{dx} * \int \mathbf{v} \, dx \right\} dx$$

L = Logarithmic functions
I = Inverse functions
A = Arithmetic functions
T = Trigonometric functions
E = Exponential function

Here u & v both are functions of variable x

Method 2 $\Rightarrow$ Example – 12

Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx.$

Solution:

$$I = \int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$$

$$= \int_0^{\log 2} \int_0^x \left(\int_0^{x+\log y} e^x e^y e^z dz \right) dy dx$$

Method 2 $\Rightarrow$ Example – 12(continue)

$$I = \int_0^{\log 2} \int_0^x e^x e^y [e^z]_{z=0}^{z=x+\log y} dy dx$$

$$= \int_0^{\log 2} \int_0^x e^x e^y [e^{x+\log y} - e^0] dy dx$$

$$= \int_0^{\log 2} \int_0^x e^x e^y [e^x e^{\log y} - 1] dy dx$$

$$= \int_0^{\log 2} \int_0^x e^x e^y [e^x y - 1] dy dx$$

$$= \int_0^{\log 2} \int_0^x [e^{2x} e^y y - e^x e^y] dy dx$$

Method 2 $\Rightarrow$ Example – 12(continue)

$$I = \int_0^{\log 2} \int_0^x [e^{2x} e^y y - e^x e^y] dy dx$$

$$I = \int_0^{\log 2} \left\{ e^{2x} \left[y e^y - e^y \right]_{y=0}^{y=x} - e^x \left[e^y \right]_{y=0}^{y=x} \right\} dx$$

$$= \int_0^{\log 2} \left\{ e^{2x} [(x e^x - e^x) - (0 - e^0)] - e^x [(e^x - e^0)] \right\} dx$$

$$\int y e^y dy$$

$$= y \int e^y dy$$

$$- \int \left[\frac{dy}{dy} \int e^y dy \right] dy$$

$$= y e^y - e^y$$

Method 2 $\Rightarrow$ Example – 12(continue)

$$I = \int_0^{\log 2} \left\{ e^{2x} [(x e^x - e^x) - (0 - e^0)] - e^x [(e^x - e^0)] \right\} dx$$

$$= \int_0^{\log 2} \left\{ [(x e^{3x} - e^{3x}) + (\cancel{e^{2x}})] - [(\cancel{e^{2x}} - e^{1x})] \right\} dx$$

$$= \int_0^{\log 2} \left\{ x e^{3x} - e^{3x} + e^x \right\} dx$$

.....

$$I = \frac{8 \log 2}{3} - \frac{19}{9}$$

$$\log(x^y) = y \log x$$

$$e^{y \log x} = e^{\log(x^y)} \\ = (x^y)$$

Method 2 $\Rightarrow$ Example – 15

Evaluate $\int_1^e \int_1^{\log y} \int_1^{e^x} (x^2 + y^2) \, dz \, dx \, dy$.

Solution: W-18(4 Mark)

$$I = \int_1^e \int_1^{\log y} \left[\int_1^{e^x} (x^2 + y^2) \, dz \right] dx \, dy$$

$$= \int_1^e \int_1^{\log y} (x^2 + y^2) [z]_{z=1}^{z=e^x} dx \, dy$$

$$= \int_1^e \int_1^{\log y} (x^2 + y^2) [e^x - 1] dx \, dy$$

$$= \int_1^e \left[\int_1^{\log y} \left[x^2(e^x - 1) + y^2(e^x - 1) \right] dx \right] dy$$

$$\int \mathbf{u} \cdot \mathbf{v} \, dx = \mathbf{u} \int \mathbf{v} \, dx - \int \left\{ \frac{d\mathbf{u}}{dx} * \int \mathbf{v} \, dx \right\} dx$$

Method 2 $\Rightarrow$ Example – 15(continue)

$$I = \int_1^e \int_1^{\log y} [x^2(e^x) - x^2 + y^2(e^x - 1)] dx dy$$

$$I = \int_1^e \left\{ \begin{aligned} &[x^2 e^x - 2x e^x + 2e^x]_{x=1}^{x=\log y} \\ &- \left[\frac{x^3}{3} \right]_{x=1}^{x=\log y} + y^2 [e^x - x]_{x=1}^{x=\log y} \end{aligned} \right\} dy$$

$$\int x^2 e^x dx$$

$$= x^2 e^x - \int [2x \cdot e^x] dx$$

$$= x^2 e^x - \left[2x \cdot e^x - \int [2 \cdot e^x] dx \right]$$

$$= x^2 e^x - 2x e^x + 2e^x$$

Method 2 $\Rightarrow$ Example – 15(continue)

$$\begin{aligned}
 I &= \int_1^e \left\{ x^2 e^x - 2 x e^x + 2 e^x - \frac{x^3}{3} + y^2 [e^x - x] \right\}_{x=1}^{x=\log y} dy \\
 &= \int_1^e \left[\left\{ (\log y)^2 e^{\log y} - 2 \log y e^{\log y} + 2 e^{\log y} - \frac{(\log y)^3}{3} + y^2 [e^{\log y} - \log y] \right\} - \left\{ (1)^2 e^1 - 2 \cdot 1 e^1 + 2 e^1 - \frac{(1)^3}{3} + y^2 [e^1 - 1] \right\} \right] dy \\
 &= \int_1^e \left[\left\{ (\log y)^2 y - 2 \log y \cdot y + 2y - \frac{(\log y)^3}{3} + y^2 [y - \log y] \right\} - \left\{ e - \cancel{2e} + \cancel{2e} - \frac{1}{3} + y^2 [e - 1] \right\} \right] dy
 \end{aligned}$$

Method 2 $\Rightarrow$ Example – 15(continue)

$$I = \int_1^e \left[\left\{ \begin{aligned} &(\log y)^2 y - 2 \log y \cdot y + 2y \\ &-\frac{(\log y)^3}{3} + y^2 [y - \log y] \end{aligned} \right\} - \left\{ e - \frac{1}{3} + y^2 [e - 1] \right\} \right] dy$$

$$= \int_1^e \left[\underbrace{y \log^2 y}_{\downarrow} - \underbrace{2 y \log y}_{\downarrow} - \underbrace{y^2 \log y}_{\downarrow} - \underbrace{\frac{\log^3 y}{3}}_{\downarrow} + \underbrace{y^3 + y^2 - ey^2 + 2y - e + \frac{1}{3}}_{\downarrow} \right] dy$$

$$\left[\frac{e^2}{4} - \frac{1}{4} \right] - \left[\frac{e^2}{2} + \frac{1}{2} \right] - \left[\frac{2e^3}{9} + \frac{1}{9} \right] - \left[2 - \frac{2e}{3} \right] + \left[\frac{-e^4}{12} + \frac{e^3}{3} + \frac{5e}{3} - \frac{23}{12} \right]$$



$$I = (-3e^4 + 4e^3 - 9e^2 + 84e - 172)/36$$

Method 2 $\Rightarrow$ Example – 16

Evaluate $\iiint_V dv$, where v is the solid region bounded by
 $1 \leq x \leq 2, 2 \leq y \leq 4$ & $2 \leq z \leq 5$.

Solution:

$$I = \iiint_V 1 \, dv = \int_{z=2}^5 \int_{y=2}^4 \left(\int_{x=1}^2 1 \, dx \right) dy \, dz$$

.....

$$I = 6$$

Method 2 $\Rightarrow$ Example – 17

Evaluate $\int_0^1 \int_0^{\sqrt{z}} \int_0^{2\pi} (r^2 \cos^2 \theta + z^2) r \, d\theta \, dr \, dz.$

Solution:

$$I = \int_0^1 \int_0^{\sqrt{z}} \int_0^{2\pi} (r^2 \cos^2 \theta + z^2) r \, d\theta \, dr \, dz$$

$$= \int_0^1 \int_0^{\sqrt{z}} \left(\int_0^{2\pi} (r^3 \cos^2 \theta + r z^2) \, d\theta \right) dr \, dz$$

Method 2 $\Rightarrow$ Example – 17(continue)

$$I = \int_0^1 \int_0^{\sqrt{z}} \int_0^{2\pi} \left(r^3 \left\{ \frac{1}{2} + \frac{\cos 2\theta}{2} \right\} + r z^2 \right) d\theta \, dr \, dz$$

$$= \int_0^1 \int_0^{\sqrt{z}} \left[r^3 \left\{ \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right\} + r z^2 \theta \right]_{\theta=0}^{\theta=2\pi} dr \, dz$$

$$I = \int_0^1 \int_0^{\sqrt{z}} [r^3 \{ \pi \} + r z^2 \{ 2\pi \}] \, dr \, dz$$

.....

$$I = \frac{\pi}{3}$$

Method 2 $\Rightarrow$ Example – 18

Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^a r^7 \sin^3 \theta \cos \theta \sin \phi \cos \phi \, dr \, d\theta \, d\phi$.

Solution:

How to Write
answer in Exam?

Method 2 $\Rightarrow$ Example - 18(continue)

$$\begin{aligned}
 \frac{80^{11}-18}{n-2} \quad I &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^a \underbrace{r^7}_{\text{}} \sin^3 \theta \cdot \cos \theta \cdot \sin \phi \cdot \cos \phi \, dr \, d\theta \, d\phi \\
 &= \int_0^{\pi/2} \int_0^{\pi/2} \underbrace{\sin^3 \theta \cdot \cos \theta}_{\text{}} \cdot \sin \phi \cdot \cos \phi \left[\frac{r^8}{8} \right]_0^a d\theta \, d\phi \\
 &= \frac{a^8}{8} \int_0^{\pi/2} \left[\frac{\sin^4 \theta}{4} \right]_0^{\pi/2} \sin \phi \cdot \cos \phi \, d\phi \\
 &= \frac{a^8}{8} \cdot \frac{1}{4} \cdot \frac{1}{2} \int_0^{\pi/2} \underbrace{\sin 2\phi}_{\text{}} \, d\phi
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{a^8}{64} \left[-\frac{\cos 2\phi}{2} \right]_0^{\pi/2} \\
 &= -\frac{a^8}{64} \left[\frac{-(-1)}{2} - \frac{1}{2} \right] \\
 &= \boxed{\frac{a^8}{64}}
 \end{aligned}$$



Double Integrals Over General Region - CC

Strip parallel to X-axis

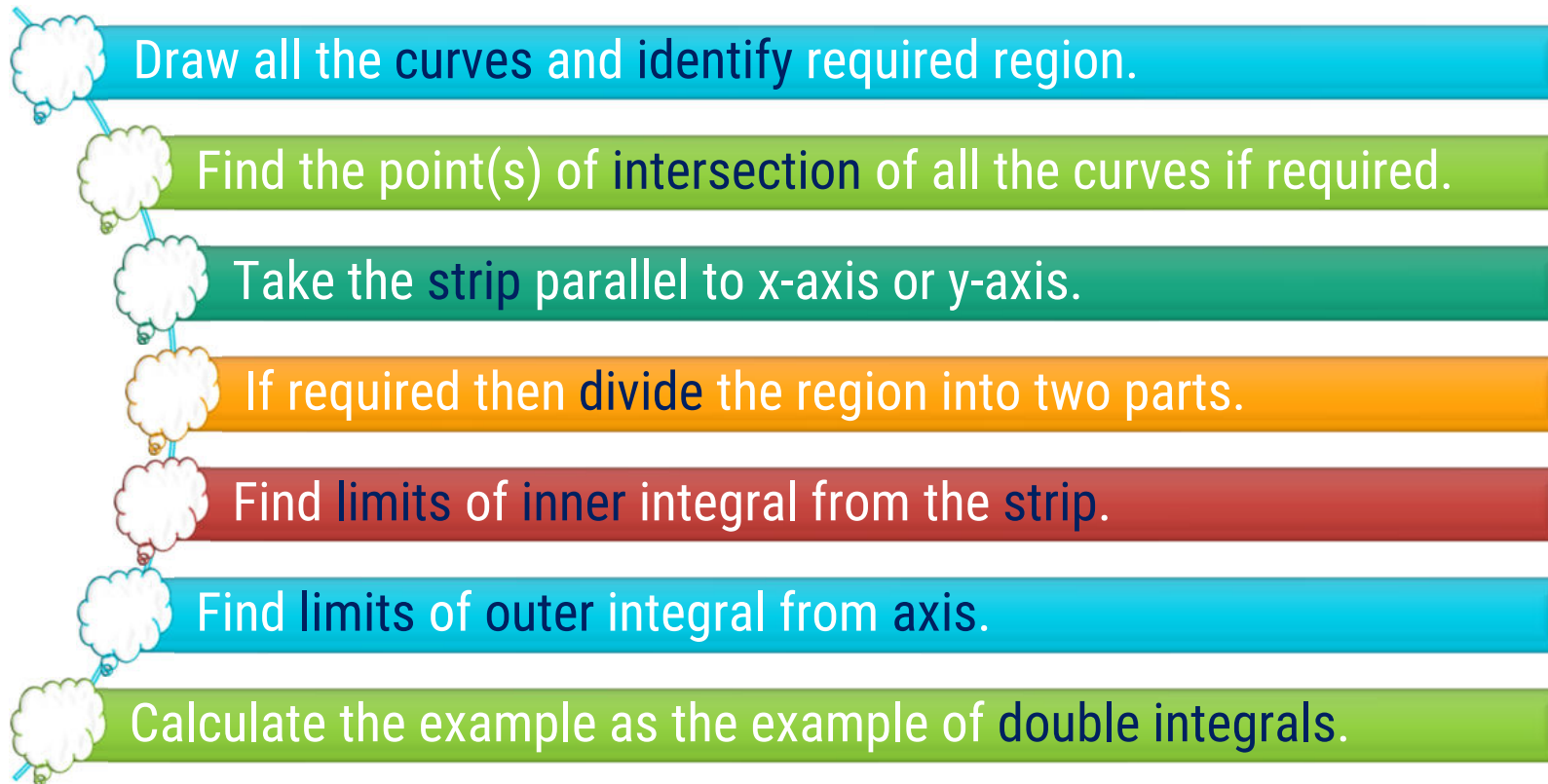
Strip parallel to Y-axis

Division of the Region



Procedure to find double integral over General Region - CC

- If we want to find the value of given double integral over general region in CC, we will follow following steps:



Strip parallel to X-axis – Limit of Inner Integral

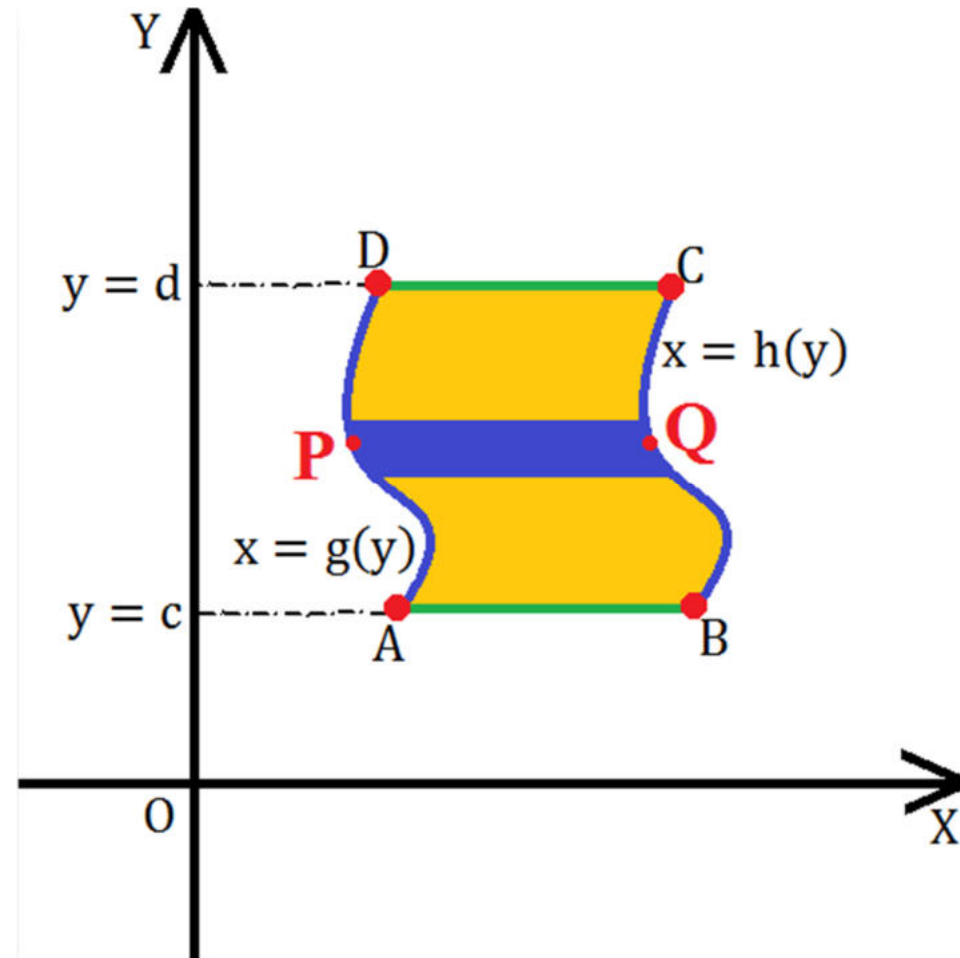
If we take a strip parallel to X-axis then

➤ Curve at **starting point** P of strip gives
lower limit of inner integral

i.e. $x = g(y)$.

➤ Curve at **end point** Q of strip gives
upper limit of inner integral

i.e. $x = h(y)$.



Strip parallel to X-axis – Limit of outer Integral

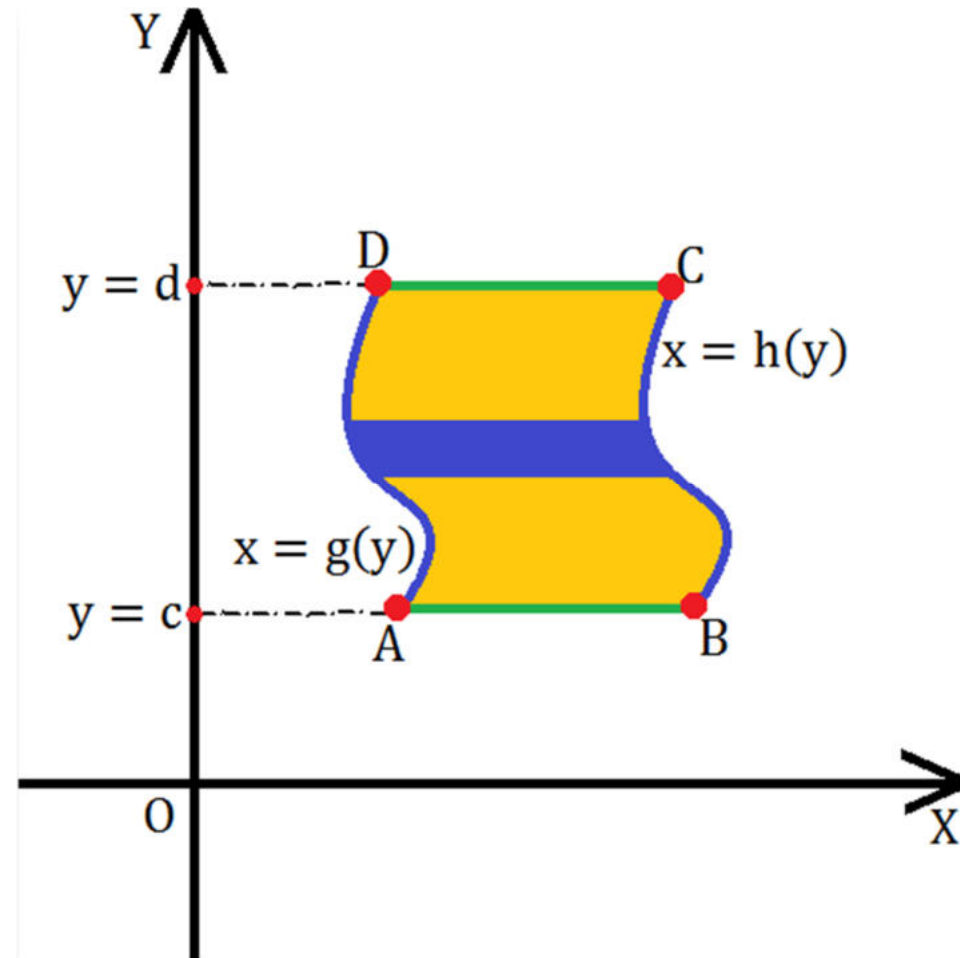
If we take a strip parallel to X-axis then

➤ Lower limit of outer integral is **starting point** of region on Y-axis

i.e. $y = c$.

➤ Upper limit of outer integral is **end point** of region on Y-axis

i.e. $y = d$.



Strip parallel to X-axis

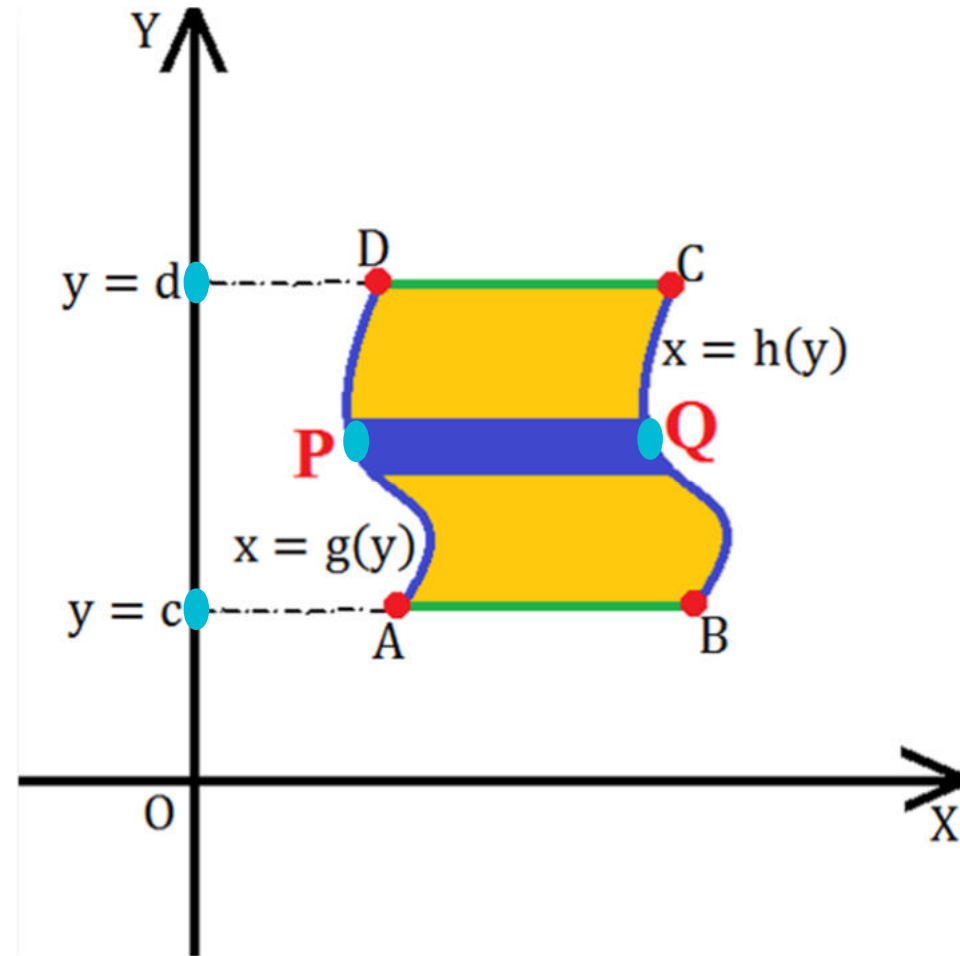
So if we take a strip parallel to **X-axis** then

✓ we have to **integrate** given function

first with respect to variable **x** and
then variable **y**.

$$\iint_R f(x, y) \, dA =$$

$$\int_{y=c}^{y=d} \left\{ \int_{x=g(y)}^{x=h(y)} f(x, y) \, dx \right\} dy$$



Strip parallel to Y-axis – Limit of Inner Integral

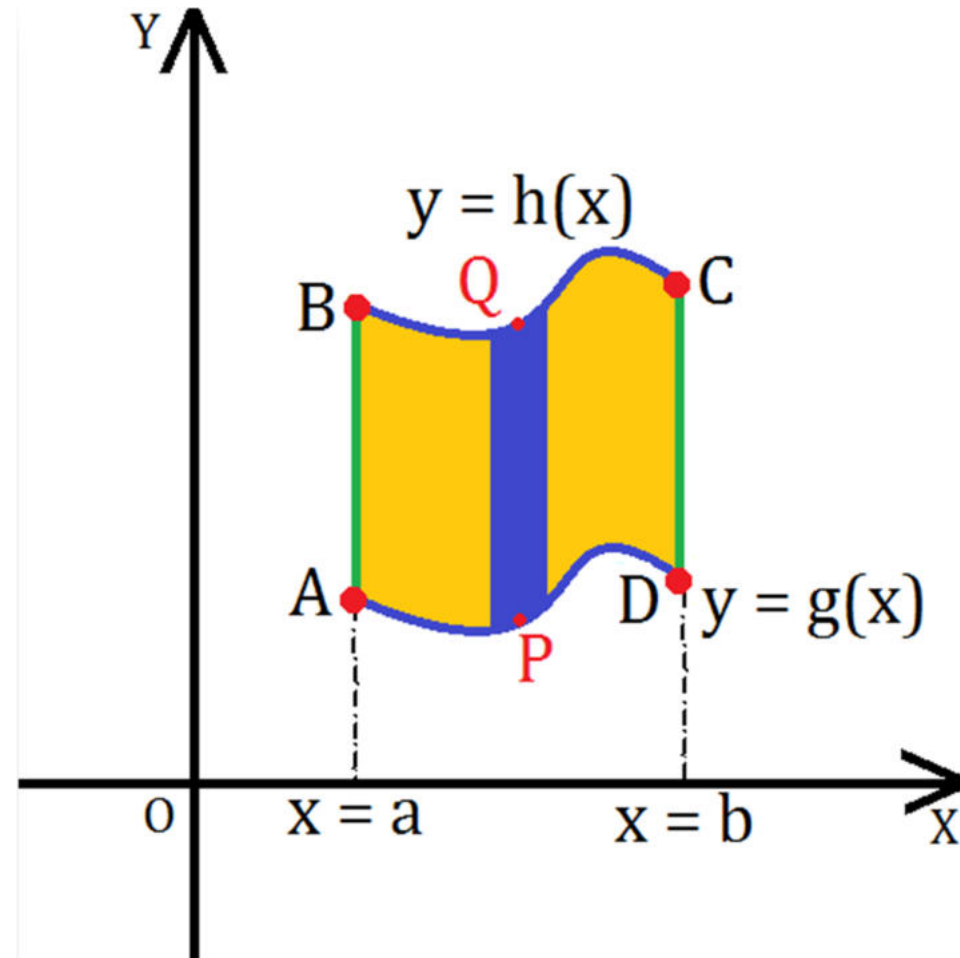
If we take a strip parallel to Y-axis then

➤ Curve at **starting point** P of strip gives
lower limit of inner integral

i.e. $y = g(x)$.

➤ Curve at **end point** Q of strip gives
upper limit of inner integral

i.e. $y = h(x)$.



Strip parallel to Y-axis – Limit of outer Integral

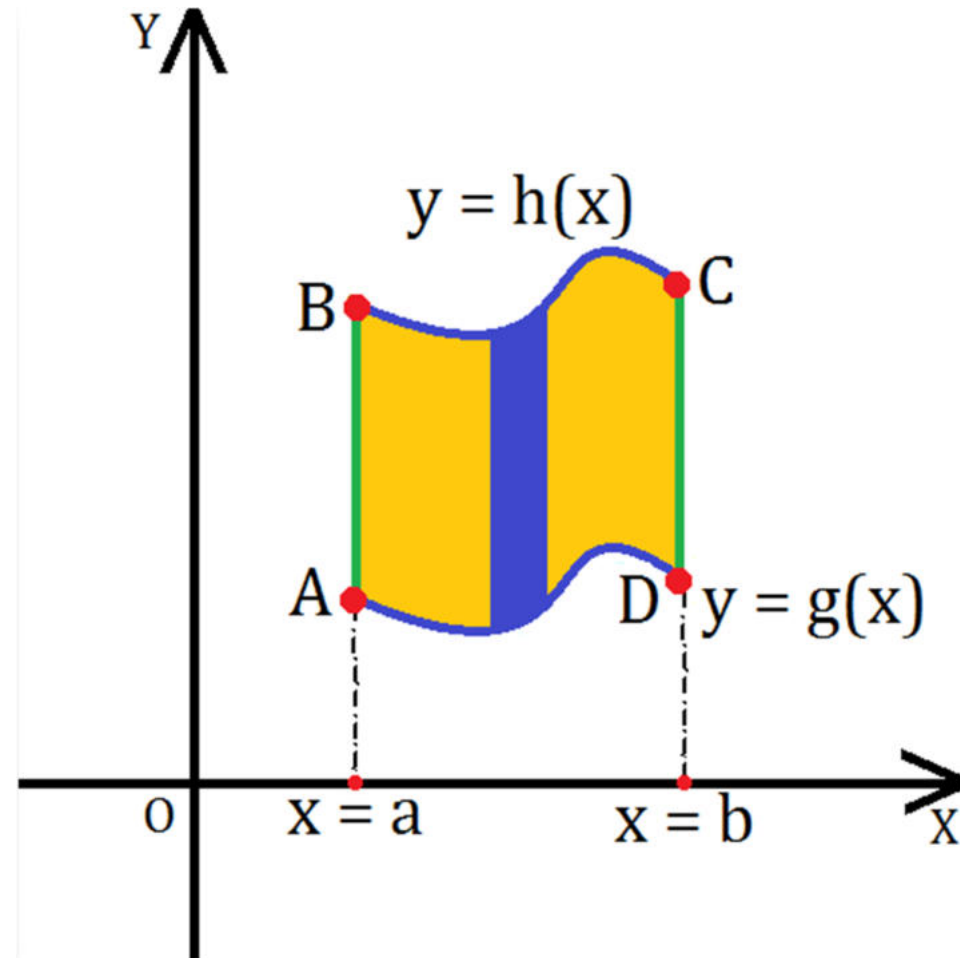
If we take a strip parallel to Y-axis then

➤ Lower limit of outer integral is starting point of region on X-axis

i.e. $x = a$.

➤ Upper limit of outer integral is end point of region on X-axis

i.e. $x = b$.



Strip parallel to Y-axis

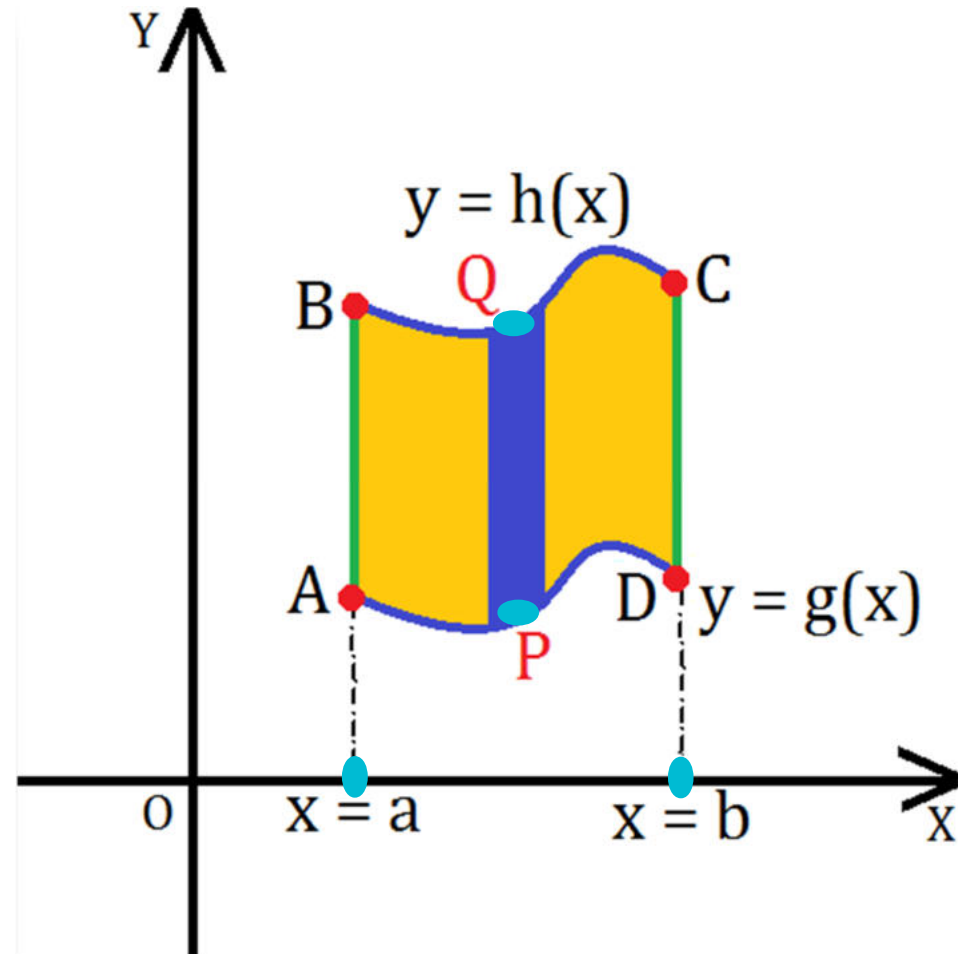
So if we take a strip parallel to Y-axis then

✓ we have to **integrate** given function

first with respect to variable **y** and
then variable **x**.

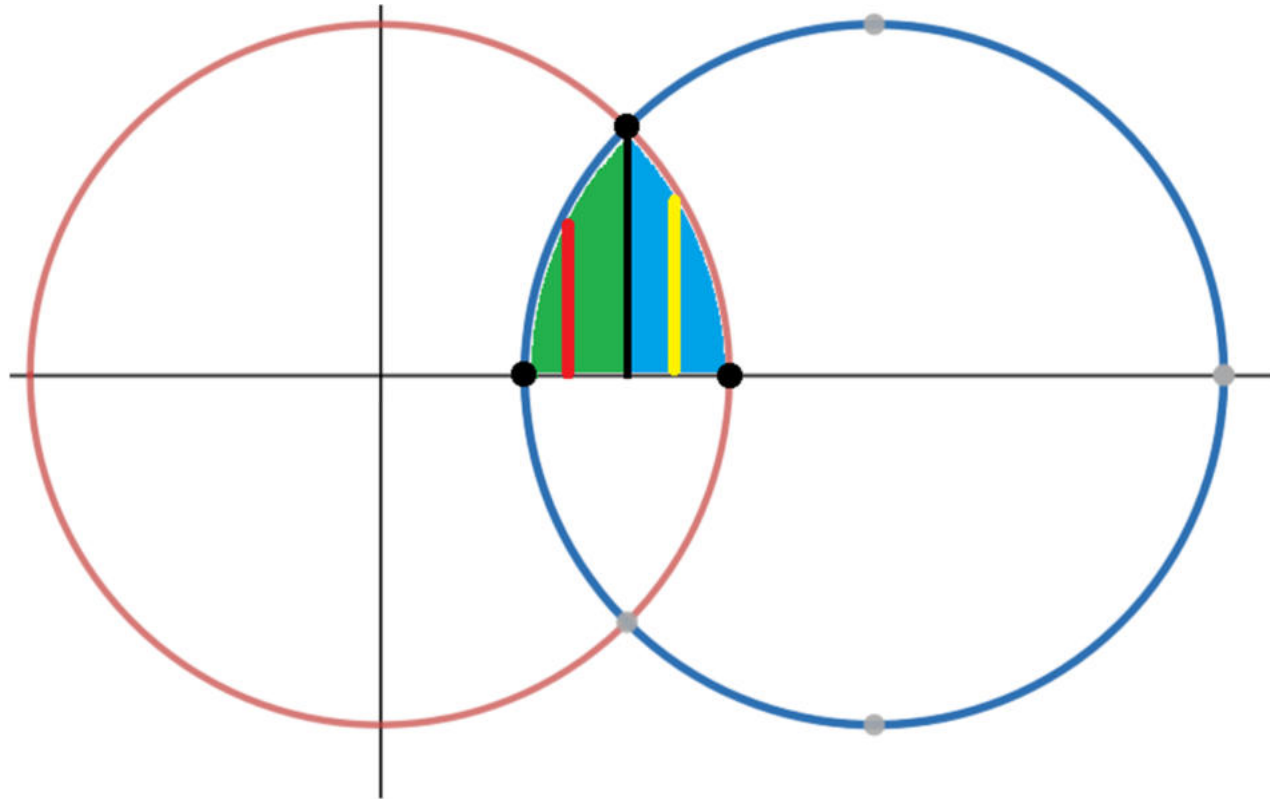
$$\iint_R f(x, y) \, dA =$$

$$\int_{x=a}^{x=b} \left\{ \int_{y=g(x)}^{y=h(x)} f(x, y) \, dy \right\} dx$$



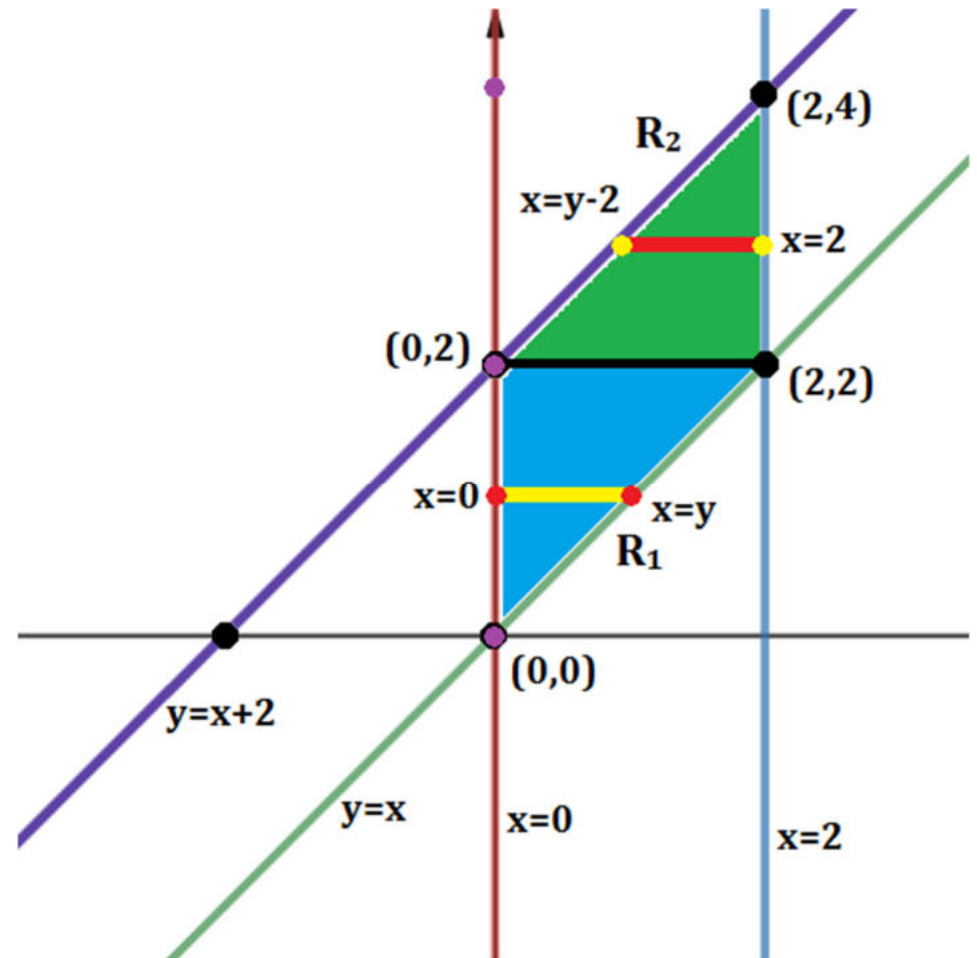
Division of the Region

- If end points of taken strip **change** the curve in given region then we have to **split** the region into more than one part.
- After splitting we have to evaluate given integral for **each** sub region.



Division of the Region

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Method 3 $\Rightarrow$ Example – 5

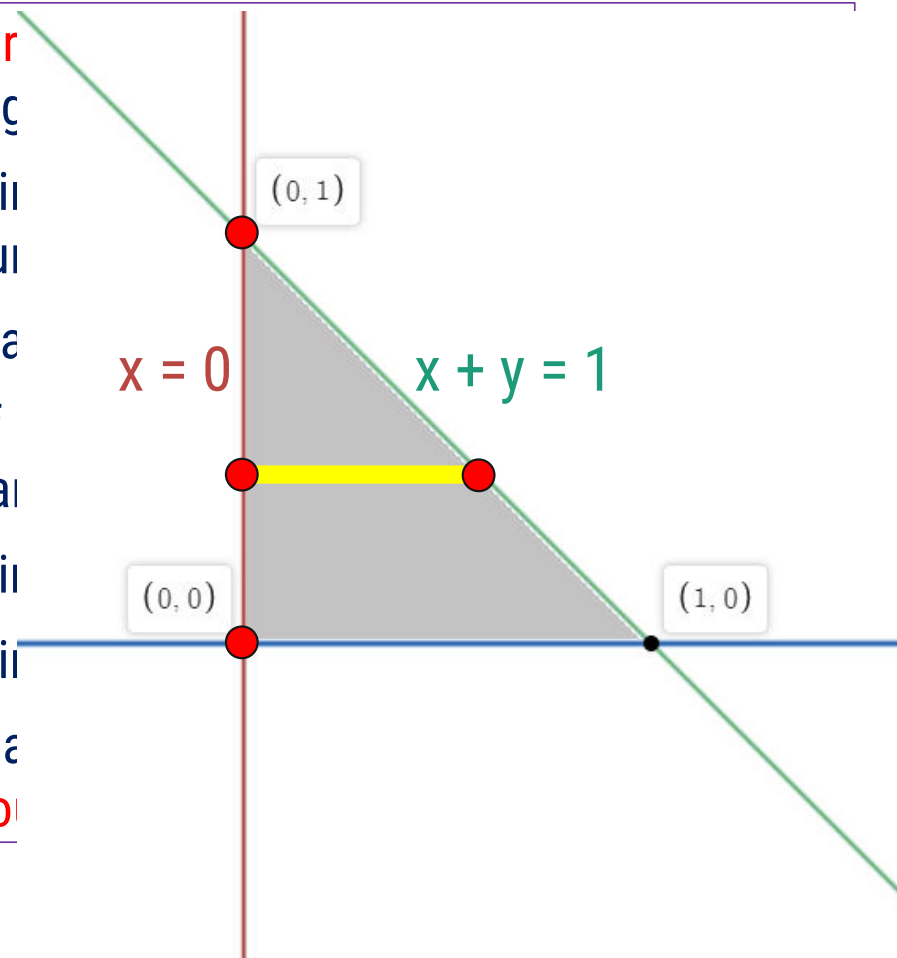
Evaluate $\iint_R e^{2x+3y} dx dy$,
where R is the triangle bounded
by $x = 0$, $y = 0$, $x + y = 1$.

Solution:

$$\iint_R e^{2x+3y} dx dy = \int_{y=0}^1 \int_{x=0}^{1-y} e^{2x+3y} dx dy$$

.....

- Dr
- reg
- Fil
- cul
- Ta
- If
- pal
- Fil
- Fil
- Ca
- do

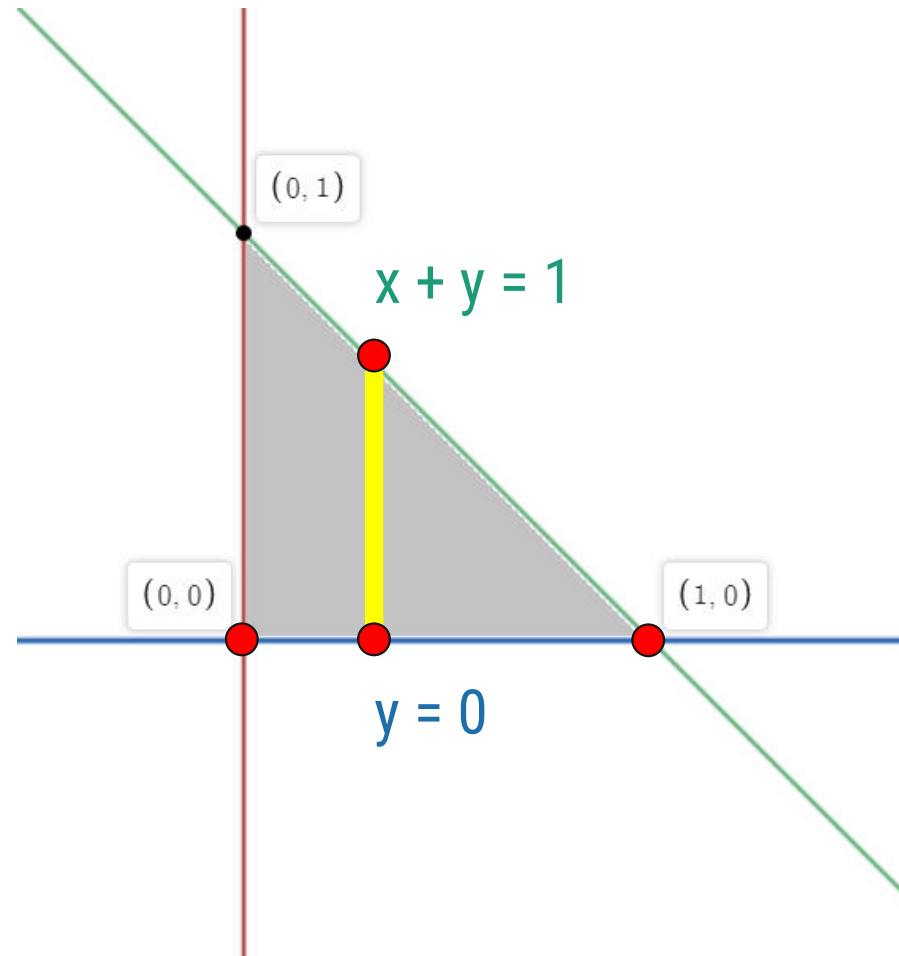


Method 3 $\Rightarrow$ Example – 5(or)

Evaluate $\iint_R e^{2x+3y} dx dy$,
where R is the triangle bounded
by $x = 0$, $y = 0$, $x + y = 1$.

Solution:

$$\iint_R e^{2x+3y} dx dy = \int_{x=0}^1 \int_{y=0}^{1-x} e^{2x+3y} dy dx$$

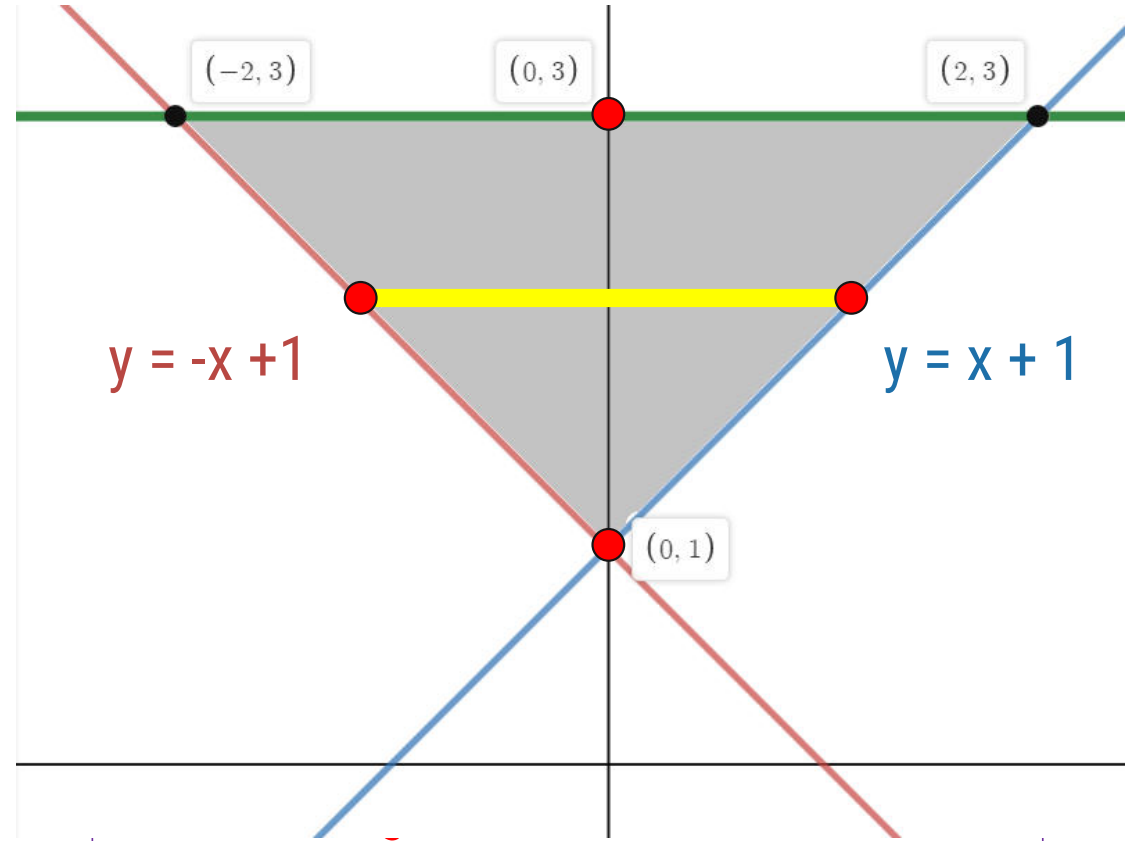


Method 3 $\Rightarrow$ Example – 8

Evaluate $\iint_R (2x - y^2) dA$; over the triangular region R enclosed between the lines $y = -x + 1$, $y = x + 1$ and $y = 3$.

Solution:

$$\iint_R (2x - y^2) dA = \int_{y=1}^3 \int_{x=1-y}^{y-1} (2x - y^2) dx dy$$

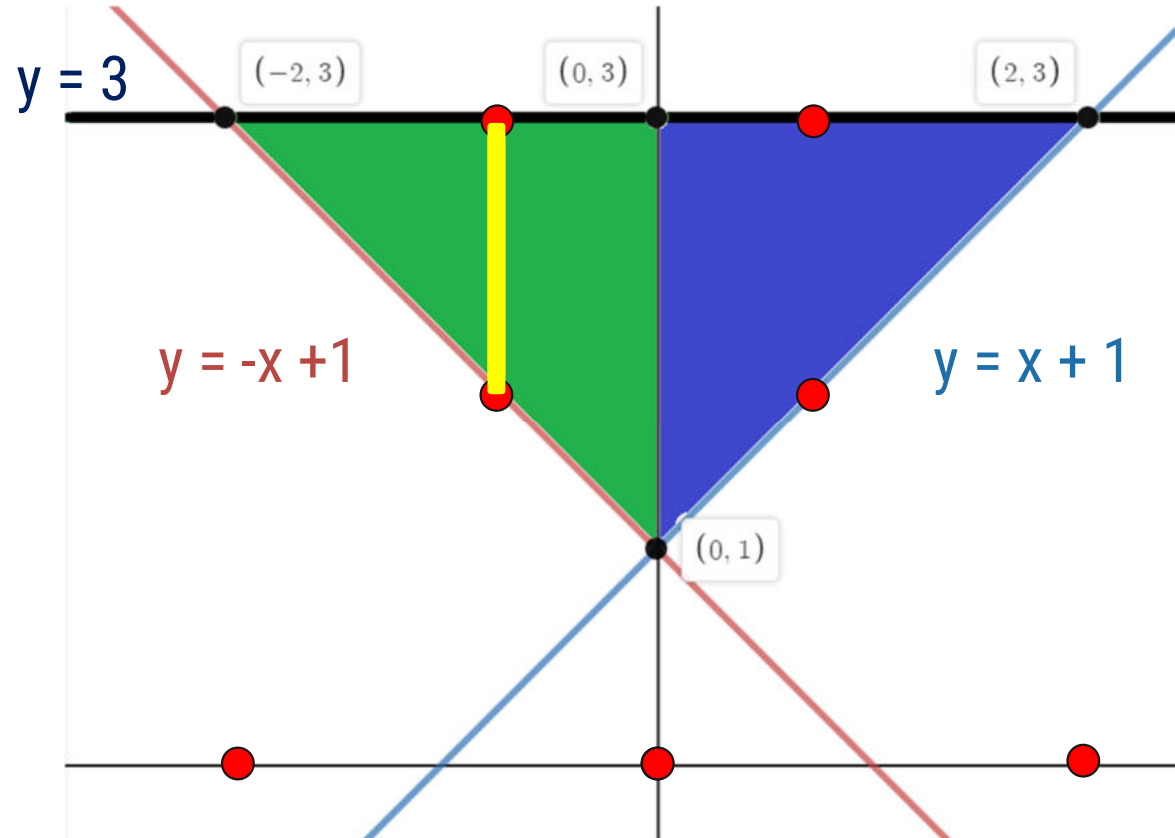


Method 3 $\Rightarrow$ Example – 8(or)

$$\iint_R (2x - y^2) dA$$

$$= \int_{x=-2}^0 \int_{y=1-x}^3 (2x - y^2) dy dx$$

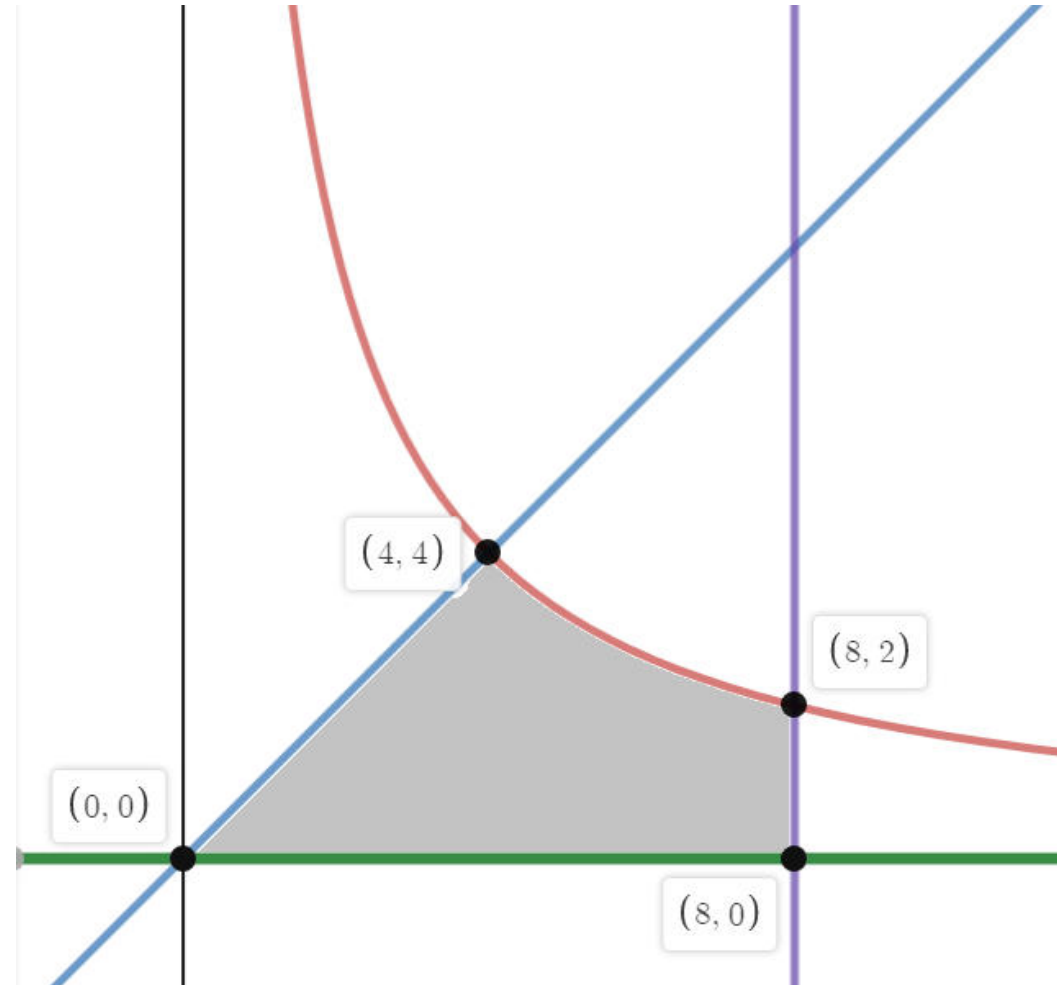
$$+ \int_{x=0}^2 \int_{y=1+x}^3 (2x - y^2) dy dx$$



Method 3 $\Rightarrow$ Example – 9

Evaluate $\iint_R x^2 \, dA$, where
 R is the region in the first quadrant bounded by the hyperbola $xy = 16$ and the lines $y = x$, $y = 0$ and $x = 8$.

Solution:



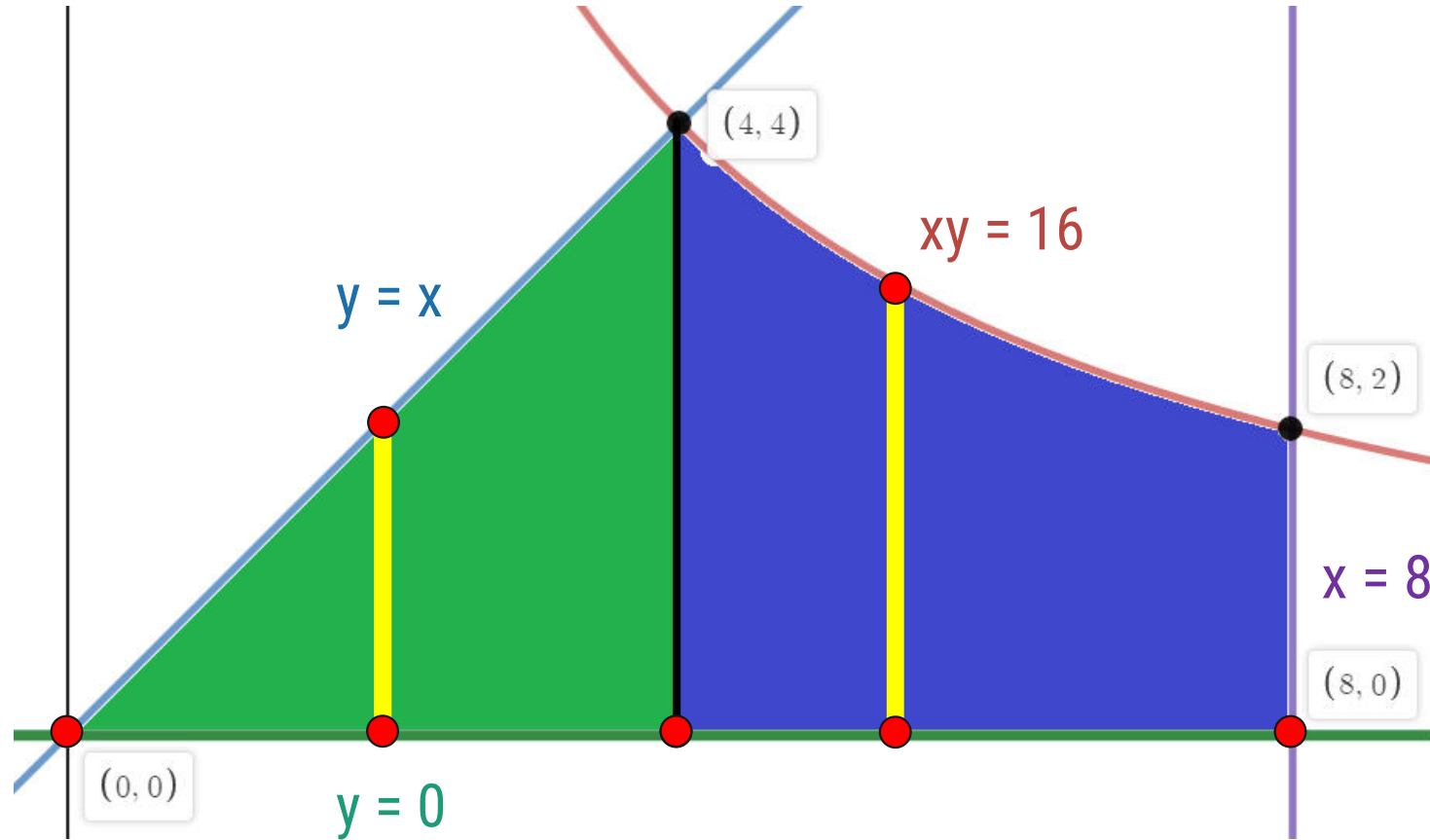
Method 3 $\Rightarrow$ Example – 9(continue)

$$\iint_R x^2 dA$$

$$= \int_{x=0}^4 \int_{y=0}^x x^2 dy dx$$

$$+ \int_{x=4}^8 \int_{y=0}^{16/x} x^2 dy dx$$

.....



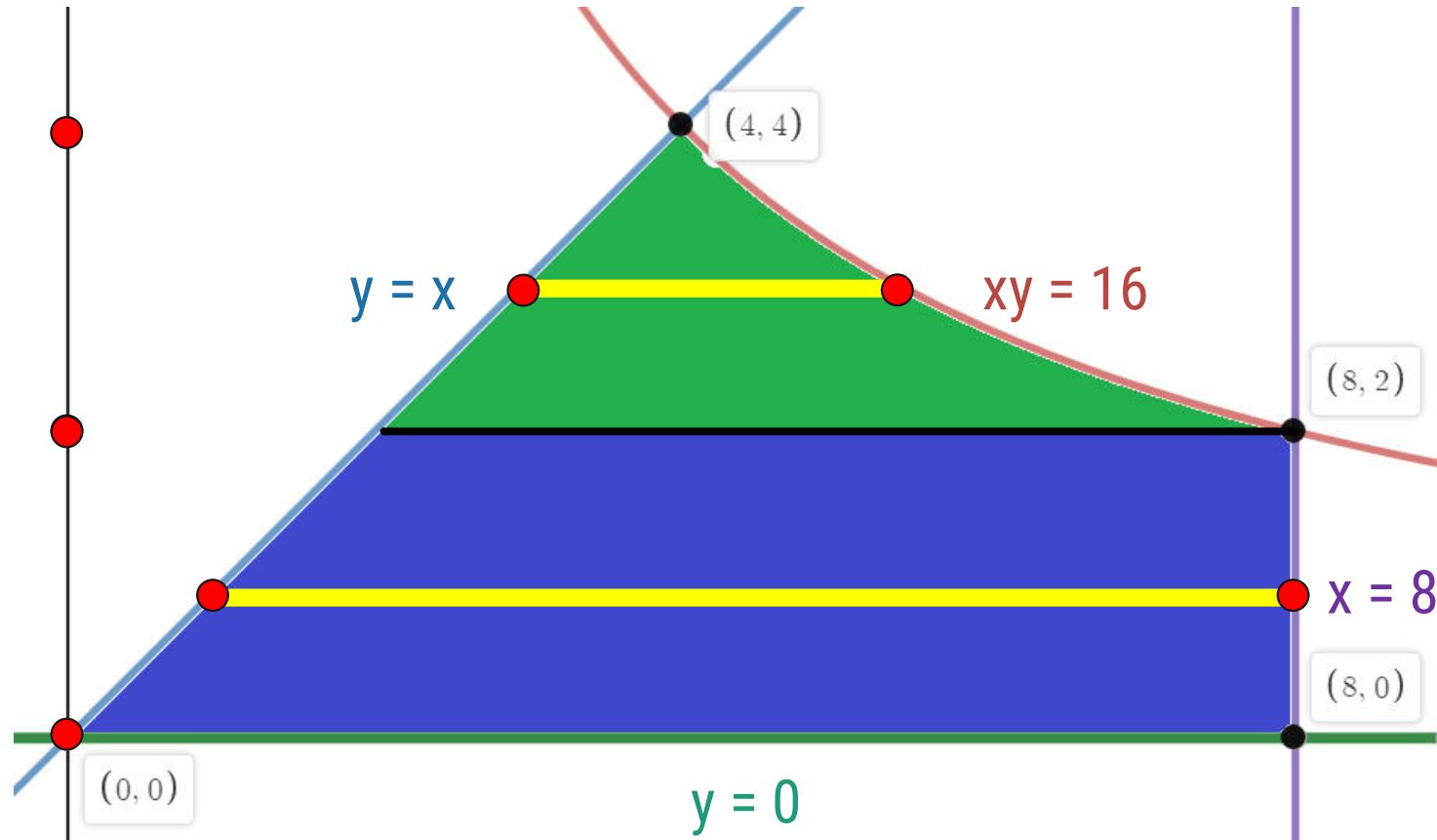
Method 3 $\Rightarrow$ Example – 9(or)

$$\iint_R x^2 dA$$

$$= \int_{y=0}^2 \int_{x=y}^8 x^2 dx dy$$

$$+ \int_{y=2}^4 \int_{x=y}^{16/y} x^2 dx dy$$

.....



Method 3 $\Rightarrow$ Example – 11

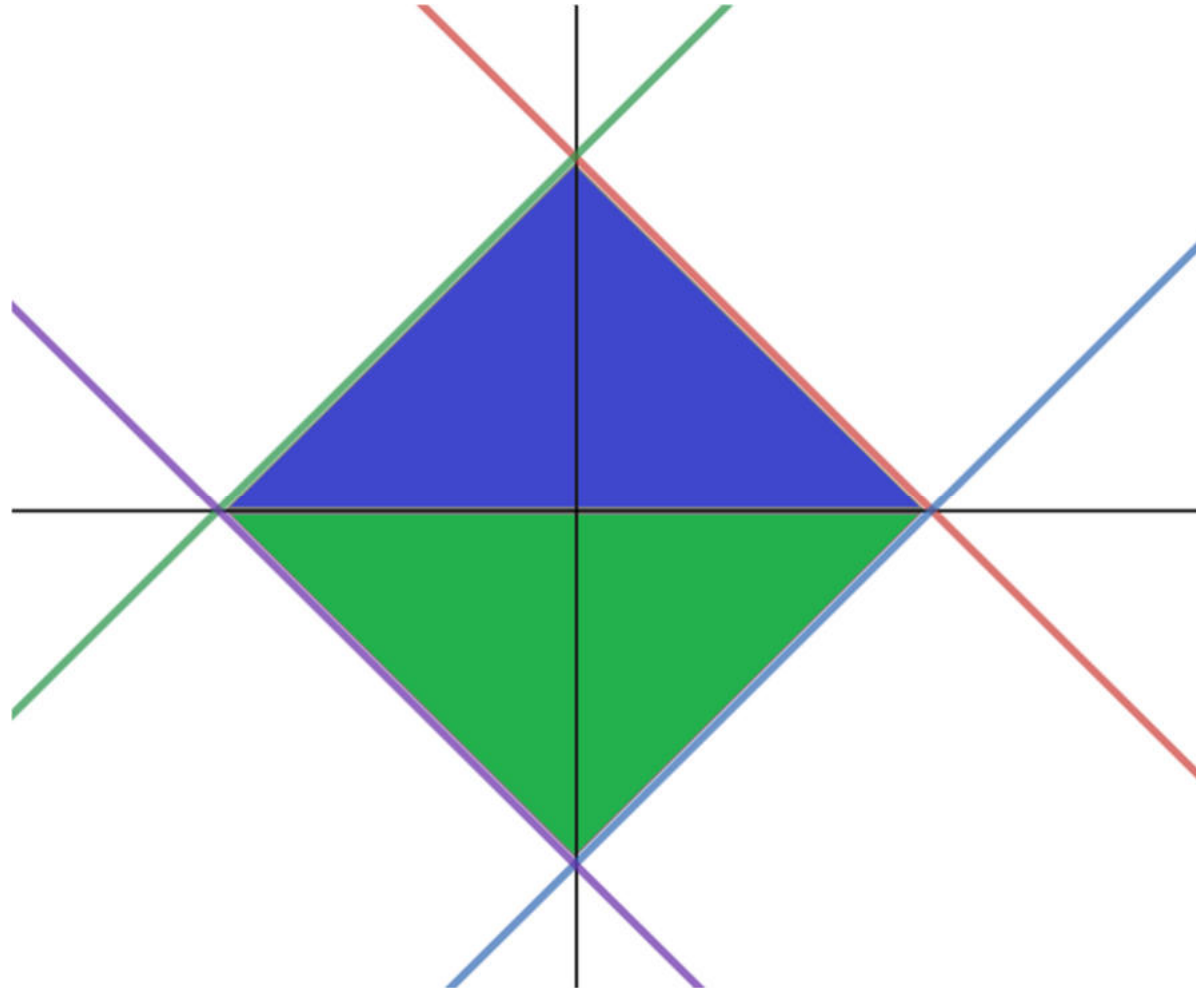
Sketch the region of integration and evaluate the integral

$$\iint_R (y - 2x^2) \, dA,$$

where R is the region inside the square $|x| + |y| = 1$.

$$\begin{cases} x + y = 1 & x - y = 1 \\ -x + y = 1 & -x - y = 1 \end{cases}$$

Solution: **S-19(4 mark)**



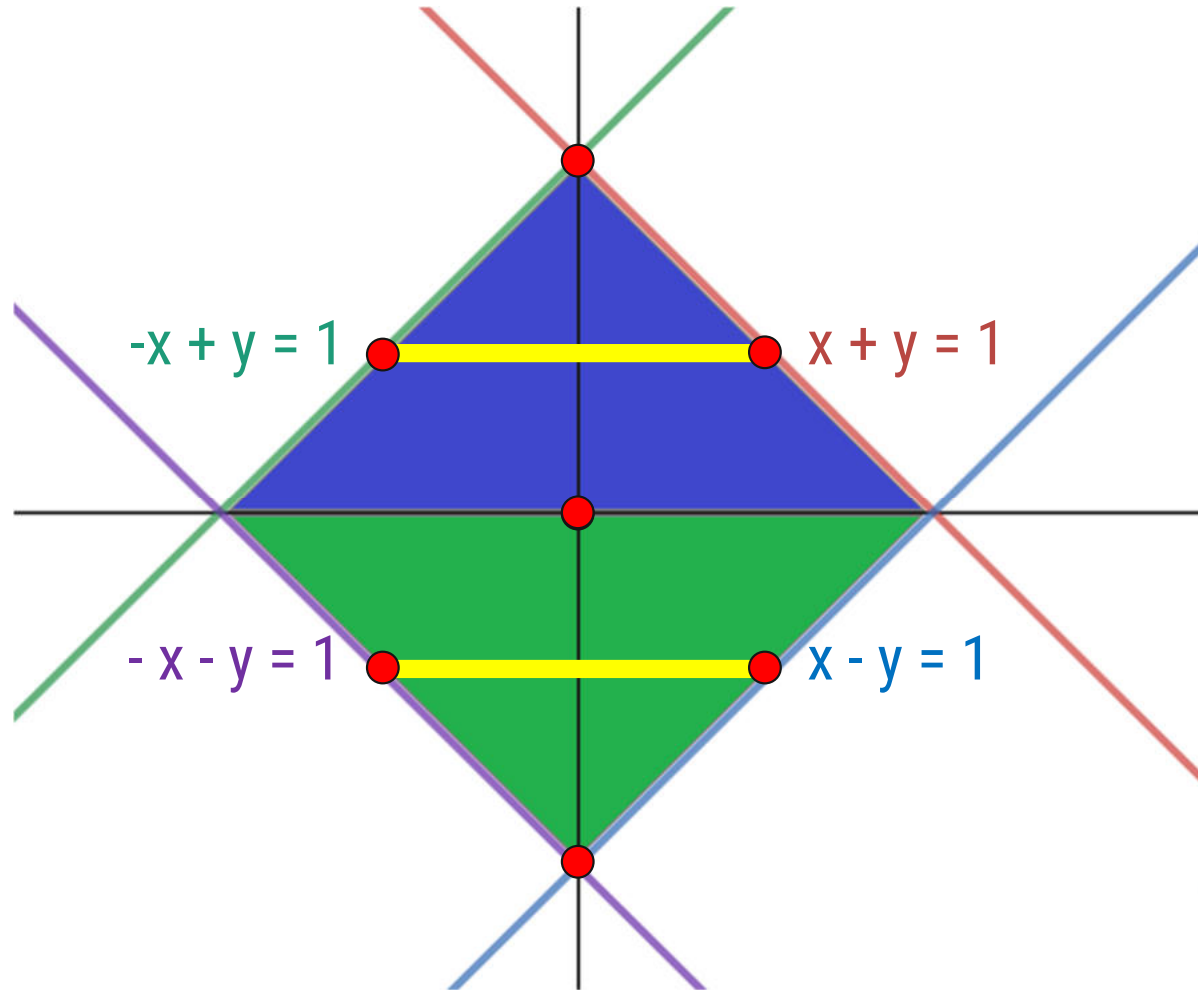
Method 3 $\Rightarrow$ Example – 11(continue)

$$\iint_R (y - 2x^2) dA$$

$$= \int_0^1 \int_{y-1}^{1-y} (y - 2x^2) dx dy$$

$$+ \int_{-1}^0 \int_{-y-1}^{1+y} (y - 2x^2) dx dy$$

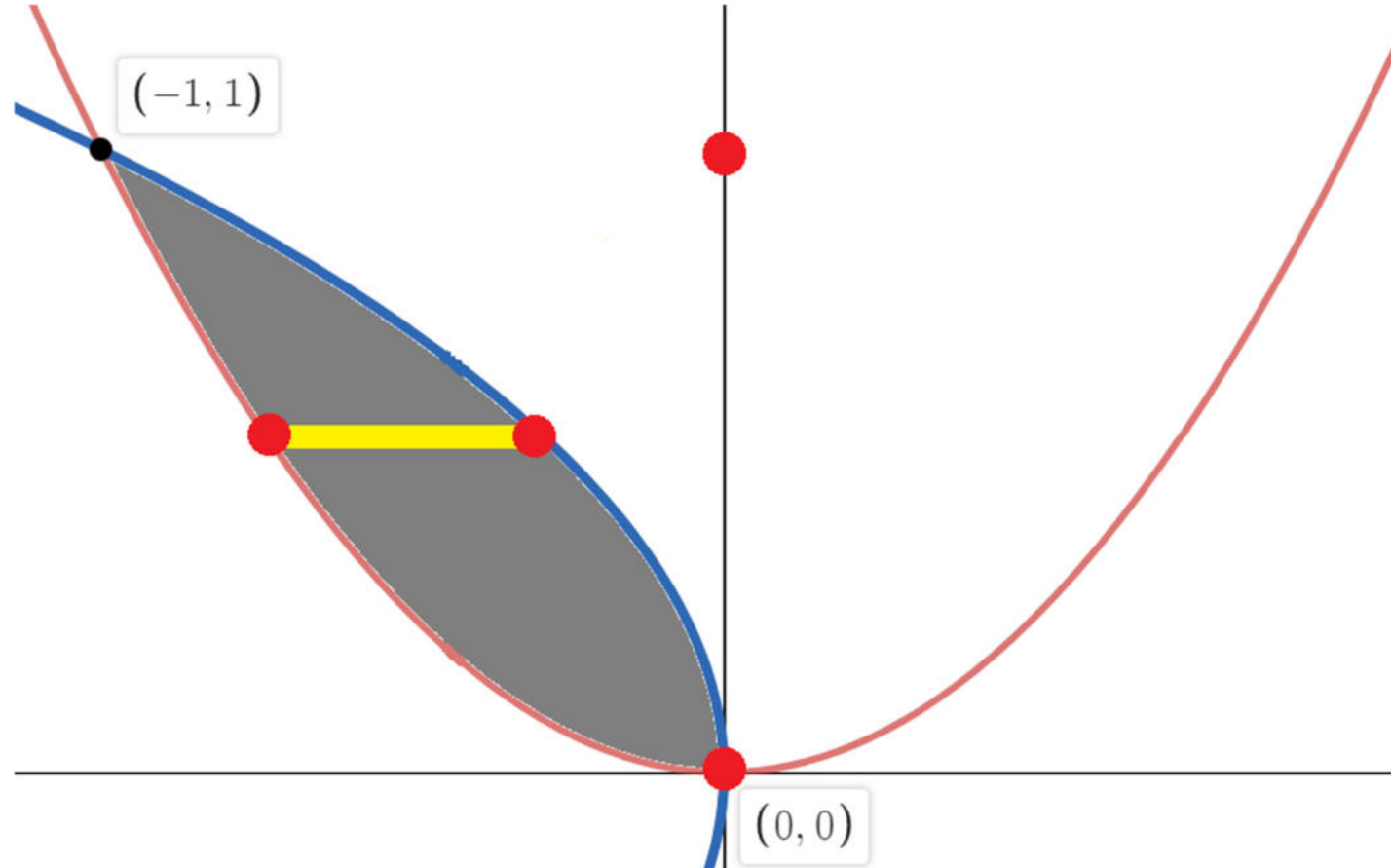
.....



Method 3 $\Rightarrow$ Example – 14

Evaluate $\iint_R (xy) \, dx \, dy$,
over the region
bounded by the
parabolas
 $x^2 = y$ and $y^2 = -x$.

Solution:

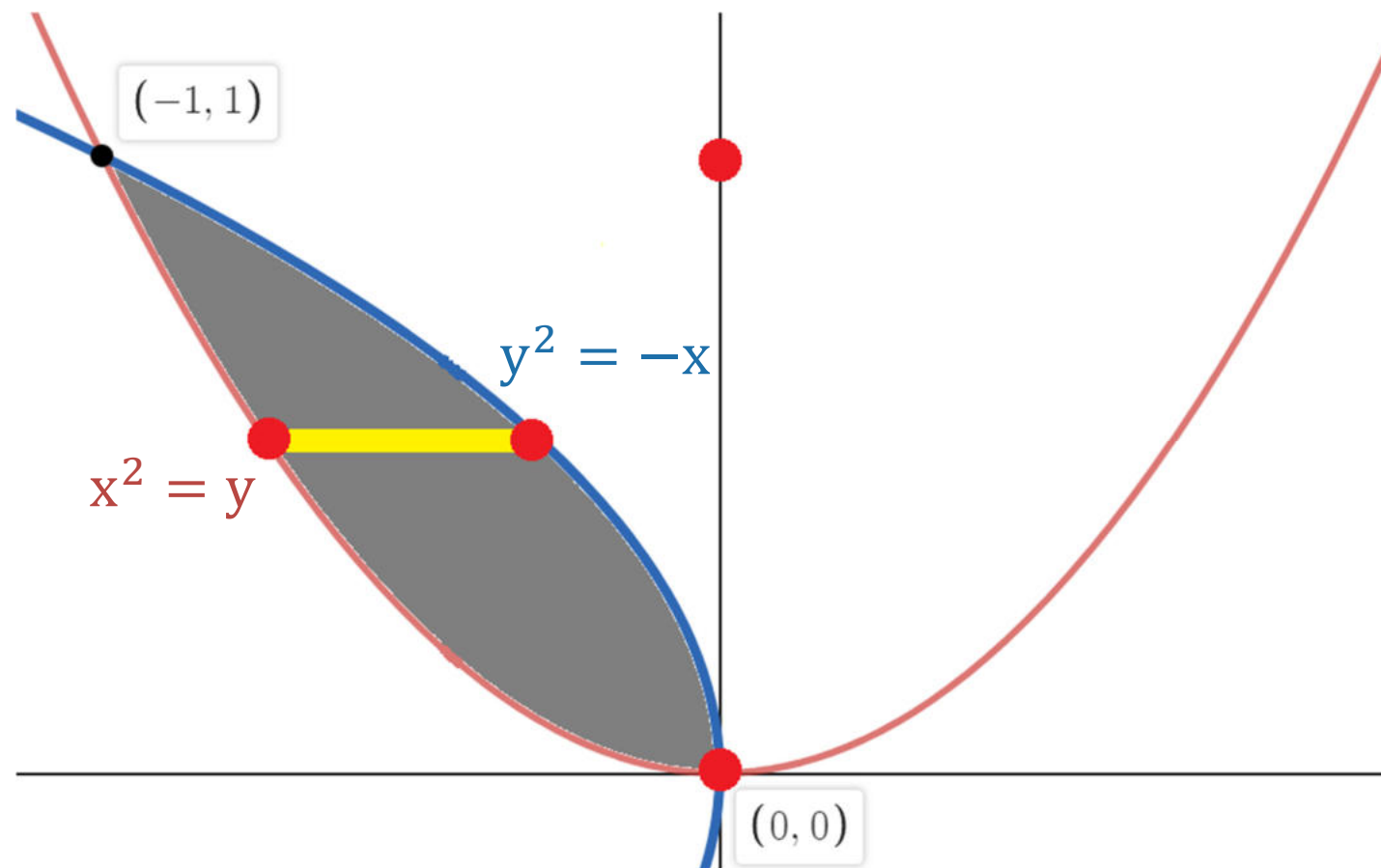


Method 3 $\Rightarrow$ Example – 14(continue)

$$\iint_R (xy) \, dx \, dy$$

$$= \int_0^1 \int_{-\sqrt{y}}^{-y^2} (xy) \, dx \, dy$$

.....



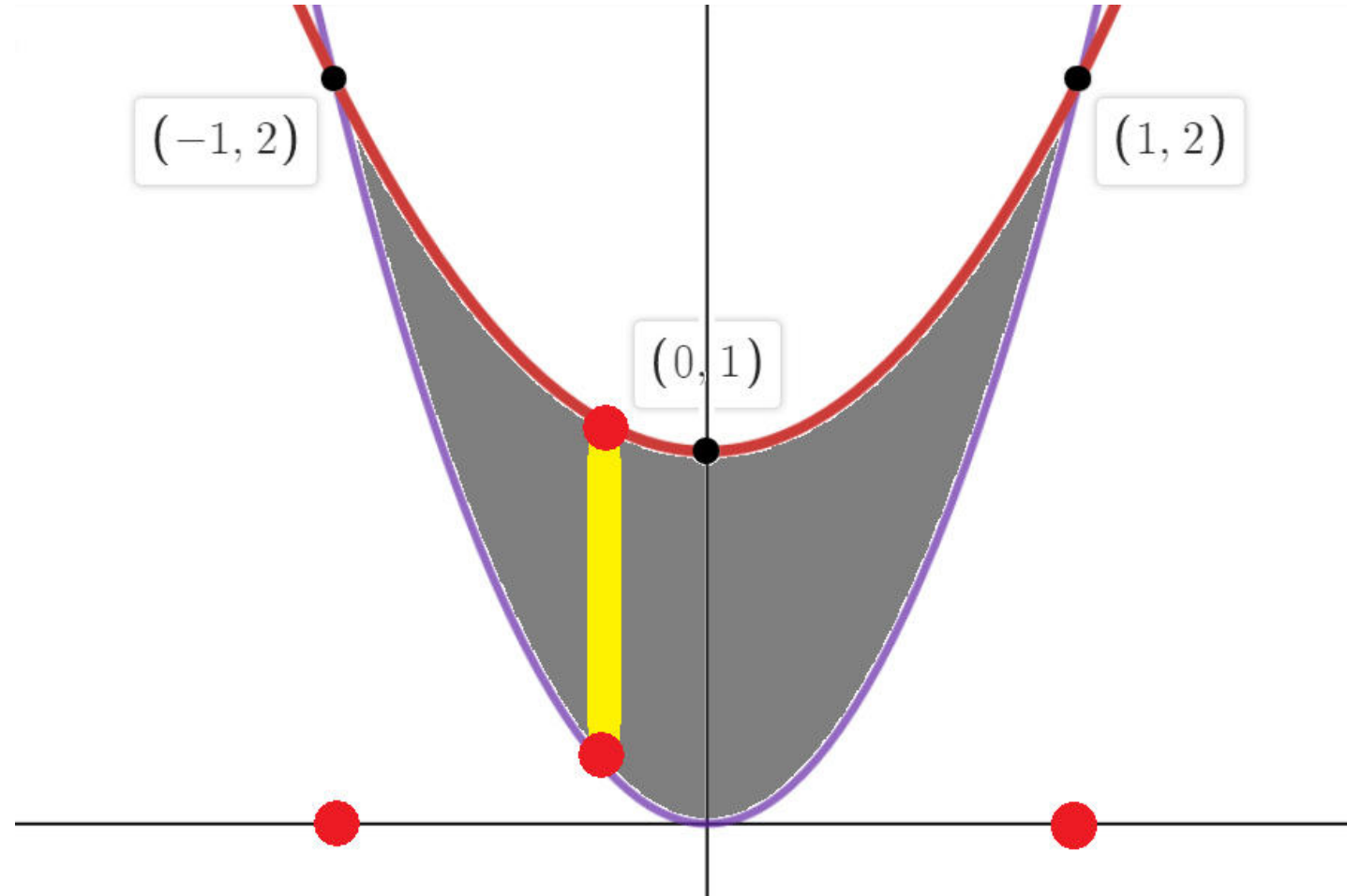
Method 3 $\Rightarrow$ Example – 19

Evaluate

$$\iint_R (x + 2y) \, dA,$$

over the region
bounded by the
parabolas $y = 2x^2$
and $y = 1 + x^2$.

Solution:

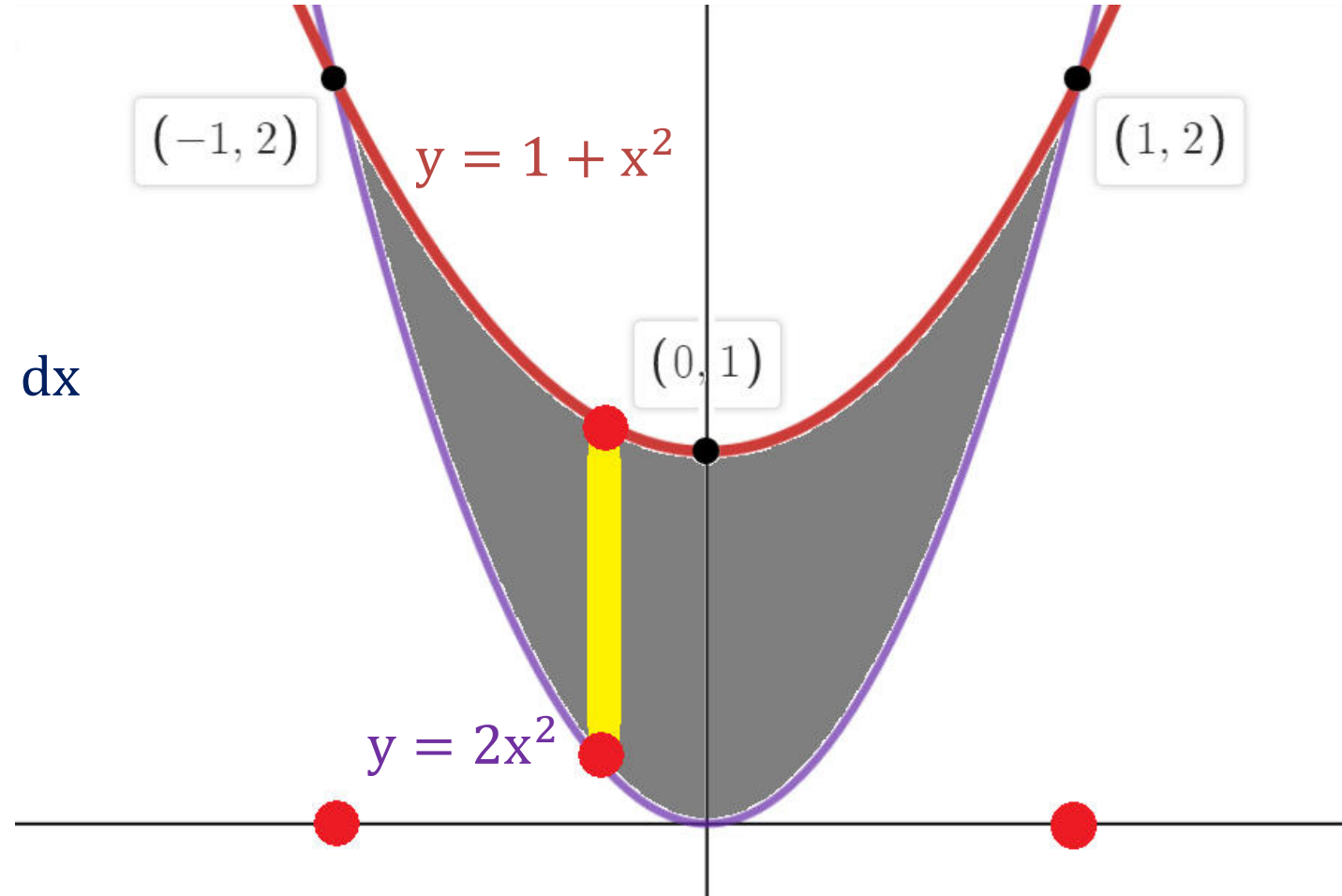


Method 3 $\Rightarrow$ Example – 19(continue)

$$\iint_R (x + 2y) \, dA$$

$$= \int_{-1}^1 \int_{2x^2}^{1+x^2} (x + 2y) \, dy \, dx$$

.....





KEEP
LEARNING
AND
ENJOY